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Backtesting and estimation error: value-at-risk overviolation rate

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Abstract
Financial institutions and regulators use value-at-risk (VaR) and related measures as a tool for financial risk management. It is therefore critical to appropriately assess the quality of VaR forecasts and reporting. The VaR estimation error creates an additional source of imprecision. We show that even an unbiased estimator of VaR is likely to produce a systematic overviolation. We then propose an adjustment to account for the issue. A Monte Carlo study illustrates the overviolation problem and the effectiveness of the adjustment. An application to Fama–French portfolios returns series highlights the need to further account for tail behavior in the data. Applying the adjustment to the normal distribution performs relatively well for a less prudential level (5% VaR), but is unable to provide enough buffer to overcome the overviolation for more prudential levels (1% or 0.5%VaR). Using the empirical distribution for more prudential levels improves risk forecasts.

Keywords Risk management · Value-at-risk · Forecasting · Backtesting · Estimation error

JEL Classification C1 · C52 · C53

1 Introduction

First created and popularized by JPMorgan RiskMetrics in the last decade of the last century as a simple measure of risk exposure, value-at-risk (VaR) has become the stan-

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dard for market risk management in the finance industry. Attracted by its simplicity and usefulness, regulators have adopted the VaR worldwide as a central measure of risk exposure. VaR defines the maximum loss on an investment over a specified time horizon at a given confidence level extreme risk. In accordance with the Basel Committee on Banking Supervision (BCBS), international financial institutions can use internal models to estimate the VaR of their market positions to comply with capital requirements. In order to assess and monitor the accuracy of VaR reporting, the BCBS in the Basel Accord (1996a) and the Amendment of Basel Accord (1996b) frame the procedure of backtesting the market VaR. These documents also provide recommendations for adjustment or "scaling up" VaR when backtesting reveals overviolation.[1]

The basic idea of backtesting is to make sure that the realized risk exposure is statistically in line with the expected one by comparing the actual losses with the reported VaRs. Overviolation which is defined as a larger than expected frequency of losses above their reported VaRs indicates an underestimation of risk. For instance, with a 99 percent confidence VaR, we should have about one percent (violation) losses exceeding their VaRs. In case there is a significant difference between the actual rate of violations and the expected one, the reported VaRs are considered not accurate. In practice, the problem is more complex and depends on how the VaR is estimated.[2] The "expected" violation rate may not be exactly the one percent used in the first place. The problem comes from the estimation error related to the parameter estimates in the VaR model or more generally to the estimation of the returns (gain/loss) distribution.[3]

Two approaches are commonly used to estimate the VaR: the nonparametric technique which makes no assumption on the distribution of returns and the parametric techniques.[4] While there are variety of alternative parametric VaR models (see Engle and Manganelli 2004; Koenker and Xiao 2006; Goureioux and Jasiak 2006, among many others.), our focus here is on the fundamental parametric specification where the VaR is the product of a standard deviation and the associated quintile of the standardized distribution. This standardized distribution can be parametric or not. In the case of nonparametric distribution, the setup is mostly semiparametric.[5]

The problem of estimating risk impact has been pointed out by some authors including Loffler (2003), who finds on portfolio credit risk that the estimation risk can increase the estimated default probability. Christoffersn and Goncalves (2005) and Escanciano and Olmo (2008) in a similar framework of VaR forecasting analyze the effect of the estimation risk on the test statistic limiting distribution for backtesting

[1] While Basel Committee on Banking Supervision is currently recommending the use of the Expected Shortfall (ES), the accuracy of the ES also relies on the accuracy of the VaR as the former is directly related to the latter. Du and Escanciano (2017) show that there is an analogy in the estimation effect on ES and VaR.

[2] Berkowitz and O'Brien (2002) examine the actual reporting of internal VaR Forecasts from six large banks and find many inaccuracies, especially in the wake of Russian default crisis when the risk in volatility estimation is greater.

[3] The VaR framework is also very useful for volatility predictions (see Amendola and Candila 2016; Chin and Lee 2018).

[4] A survey of benchmark parametric methods in the recent literature can be found in Nieto and Ruiz (2016), while Sener et al. (2012) rank the predictive performances of many VaR estimation methods.

[5] Scheller and Auer (2018) show that different VaR estimation techniques tend to deliver similar weights for asset allocation decisions.

procedures and find that ignoring estimation error may lead to misleading conclusions. Figlewski (2003) uses simulations to examine the effect of estimation errors on the VaR. Kerkhof et al. (2010) address both estimation and misspecification risks in a general framework. Alexander and Sarabia (2012) deal with the estimation risk issue for quantile risk measures in general and for the VaR especially. Other interesting assessments of the VaR are proposed by Berkowitz et al. (2011), Christoffersn and Pelletier (2004), Jorion (2002), O'Brien and Szerszen (2017), and Perignon (2008). We focus on estimation risk mainly related to the estimation of the standard error used for VaR and Expected Shortfall.

Improving the testing is one issue, while adjusting the VaR estimate is another one which can be very important in practice. Boucher et al. (2014) propose a general framework for an empirical adjustment to satisfy different backtesting criteria. Their adjustment approach is entirely nonparametric and requires very large sample to have fairly good results. Our work is closely related to the latter, but we focus on the overviolation issue which arises when sample size is relatively small.

In this paper, we show that in common parametric models used to estimate the VaR, even when the estimator of the VaR is unbiased, there is a systematic overviolation, especially when the estimation of VaR parameters depends on a relatively short window of observations. The overviolation is due to the estimation error and increases with its magnitude. We then provide simple adjustments to account for it. Based on the model used for estimation, we propose two ways to account for estimation risk in the backtesting procedure. One approach is to derive the implied expected violation rate and test whether it statistically matches the actual rate of violation. Another way to account for estimation error is to adjust the VaR estimates, so that expected and actual rates are in line.

To illustrate the overviolation phenomenon, we design a Monte Carlo study, in which the true data generating process is known. The study illustrates the overviolation pattern and the effectiveness of the proposed adjustments. With a rolling window estimation of the variance used for VaR forecast, the overviolation is larger as the window is smaller. The adjustment scale factor also increases for more prudential levels of VaR estimates. Our main focus is the MA estimation which is often used with limited data. Our results suggest that the overviolation rate could lead to a scaling factor by the regulators. The adjustment in most cases will bring the rate to the acceptable range.

Application to actual data highlights the need to further account for tail behavior. Under the normal distribution, the adjustment performs relatively well for a less prudential level (5% VaR), but is less effective for more prudential levels (1% or 0.5% VaR). The empirical distribution improves risk forecasts for more prudential levels. The rest of the paper is organized as follows. Section 2 presents the overviolation issue with theoretical results establishing the phenomenon. Common backtesting procedures are described in Sect. 3, while Sect. 4 provides the adjustment to account for overviolation. A Monte Carlo study is performed in Sect. 5 to illustrate the issue. Section 6 provides an empirical analysis of the overviolation and adjustment procedures, while Sect. 7 concludes.

2 Estimation error: sample overviolation issue

Our focus is to provide a clear understanding of the problem of overviolation due to estimation risk. Therefore, we will start by the framework of a parametric model with the common assumption of normality for gain/loss distribution. Furthermore, by assuming that the distribution has a zero mean (which can be achieved by subtracting the sample mean), the VaR at the level α is related to the standard deviation, σ_t by the simple relation

$$VaR_{\alpha,t} = \sigma_t z_\alpha$$

where $z_\alpha = -\Phi^{-1}(\alpha)$ and Φ is the cumulative distribution function of the standard normal distribution. We should notice that $1-\alpha$ is the level of confidence which is commonly large with values ranges of (95%, 99%, 99.5%,...). In other words, α is small enough to be lower than 50%, so that $z_\alpha = \Phi^{-1}(1-\alpha)$ is positive.

The estimation of σ_t is usually done using available past data. Let us assume that a rolling window of m periods of most recent observations is used to estimate the sample standard deviation

$$\widehat{\sigma}_t = \sqrt{\frac{\sum_{\tau=t-m}^{t-1} \left(X_\tau - \overline{X}\right)^2}{m-1}}$$

where $\overline{X}$ is the sample estimate of the mean and X_τ, the gain/loss for period τ (1 day or 10 days) is assumed to be stationary with a normal distribution with standard deviation σ. Assuming that they are independent, the exact distribution of the sample standard deviation is known

$$\widehat{\sigma}_t \overset{d}{=} \sigma\sqrt{\frac{\chi^2_{m-1}}{m-1}}$$

where χ^2_{m-1} is a Chi-squared distribution with $m-1$ degrees of freedom. The accuracy of a VaR estimate is based on the principle that percentage of violation should be α, the theoretical probability of exceeding the VaR_α. For each time period $t = 1, \ldots T$ with a value-at-risk $VaR_{\alpha,t}$, we should expect the proportion of times when the losses exceed (violate) their $VaR_{\alpha,t}$ to be α. However, due to the estimation error, this is not necessarily the case. We show below why the proportion of violations will theoretically exceed α, even in the presence of an unbiased estimate[6] of σ_t. Let $I_{\alpha,t} = I\left(X_t \leq -\widehat{VaR}_{\alpha,t}\right)$ be the indicator of violation taking the value 1 for a violation and 0 otherwise. Note that $\widehat{VaR}_{\alpha,t} = \widehat{\sigma}_t z_\alpha$ is the estimate of the value-at-risk at time t.

[6] Figure 1 provides a simple illustration of the overviolation issue with an unbiased estimator of the VaR.

The violation rate is given by

$$r_\alpha = \frac{1}{T}\sum_{t=1}^{T} I_{\alpha,t}$$

This rate is the rate when comparing the actual losses to VaR forecasts. The following proposition compares the expected theoretical rate of violation to α, the rate set in advance.

Proposition 1 *The expected theoretical rate of violation is*

$$E\left(r_\alpha\right) = E\left[\Phi\left(-\frac{\widehat{\sigma}_t}{\sigma} z_\alpha\right)\right] \geq \alpha$$

Proof See "Appendix"

This result states that while we may expect the violation rate to be α, it should theoretically exceed that level. The whole intuition is based on the fact that around the true value of the $-VaR_\alpha$, the cumulative distribution function (cdf) is convex (which is equivalent to say that the density function is increasing).[7] As illustrated in Fig. 1, when we move to the right, the area increases more than it decreases when we move to the left. In other words, the error adds more probability when we underestimate the VaR than it reduces the probability when we overestimate the VaR. The overviolation can be generalized to any unbiased estimate of the VaR as follows. □

Proposition 2 *If $\widehat{\upsilon}_t$ is an unbiased estimate of the VaR, and F_x the cdf of X_t is a convex[8] function around $-VaR_\alpha$, then*

$$E\left(r_\alpha\right) = E\left(F_x\left(\widehat{\upsilon}_t\right)\right) \geq \alpha$$

Proof See "Appendix"

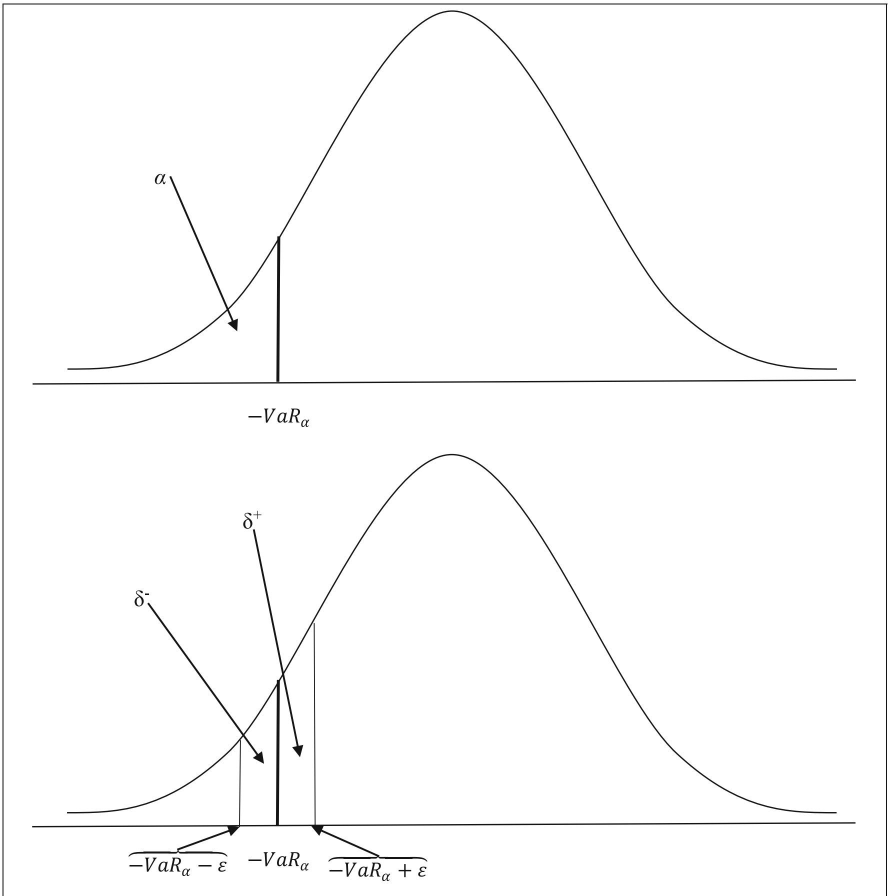
We can notice that the assumption on the cdf is very natural as it means that the density function is increasing around $(-VaR_\alpha)$. Therefore, the general result is even more interesting by stating that the problem is potentially present for any common distribution whether parametric or nonparametric. □

3 Common backtesting procedures

We present below two common backtesting procedures (unconditional and conditional coverage). The first one introduced by Kupiec (1995) tests the accuracy of violation

7 We should notice that, for a normal distribution, $-VaR_\alpha$ is always in the increasing region of the density function (left side). Therefore, stating that the cdf is convex in the area of interest is natural.

8 The convexity assumption for the cdf is very general as most density functions (derivatives of the cdf) are increasing in the left tail of the distribution. Although we do not formally prove that, even if the density is not an increasing monotone function over the relevant range of values, the trend will most likely be upward and the result will more often hold than not. Nevertheless, in some exceptional cases where in the region of interest, the distribution is mostly decreasing, the result may not hold. However, we think that those cases will be exceptions rather than the rule.



Consider an estimator of the level α value-at-risk (VaR_α) with the below distribution:

$$\hat{v} = \begin{cases} VaR_\alpha - \varepsilon, & \text{with Probability } \frac{1}{4} \\ VaR_\alpha, & \text{with Probability } \frac{1}{2}, \\ VaR_\alpha + \varepsilon, & \text{with Probability } \frac{1}{4} \end{cases}$$

Since $E(\hat{v}) = VaR_\alpha$, $\hat{v}$ is an unbiased estimator of VaR_α. However, by denoting $\hat{\alpha} \equiv \Pr(X < \hat{v})$, the violation rate obtained from the estimate, we have that $E(\hat{\alpha}) = E(\Pr(X < \hat{v})) = \alpha + \frac{\delta^+ - \delta^-}{4} > \alpha$.

Fig. 1 Illustration of the overviolation mechanism for an unbiased estimator of the value-at-risk

probability assuming independence between successive violations, while conditional coverage incorporates the possibility of dependence (see Christoffersen 2003).

3.1 Unconditional coverage

For a well-specified risk model, the time series $\{I_t\}_{t=1,\ldots T}$ should be time independent and identically distributed as Bernoulli with the probability α for 1. So we test the null hypothesis

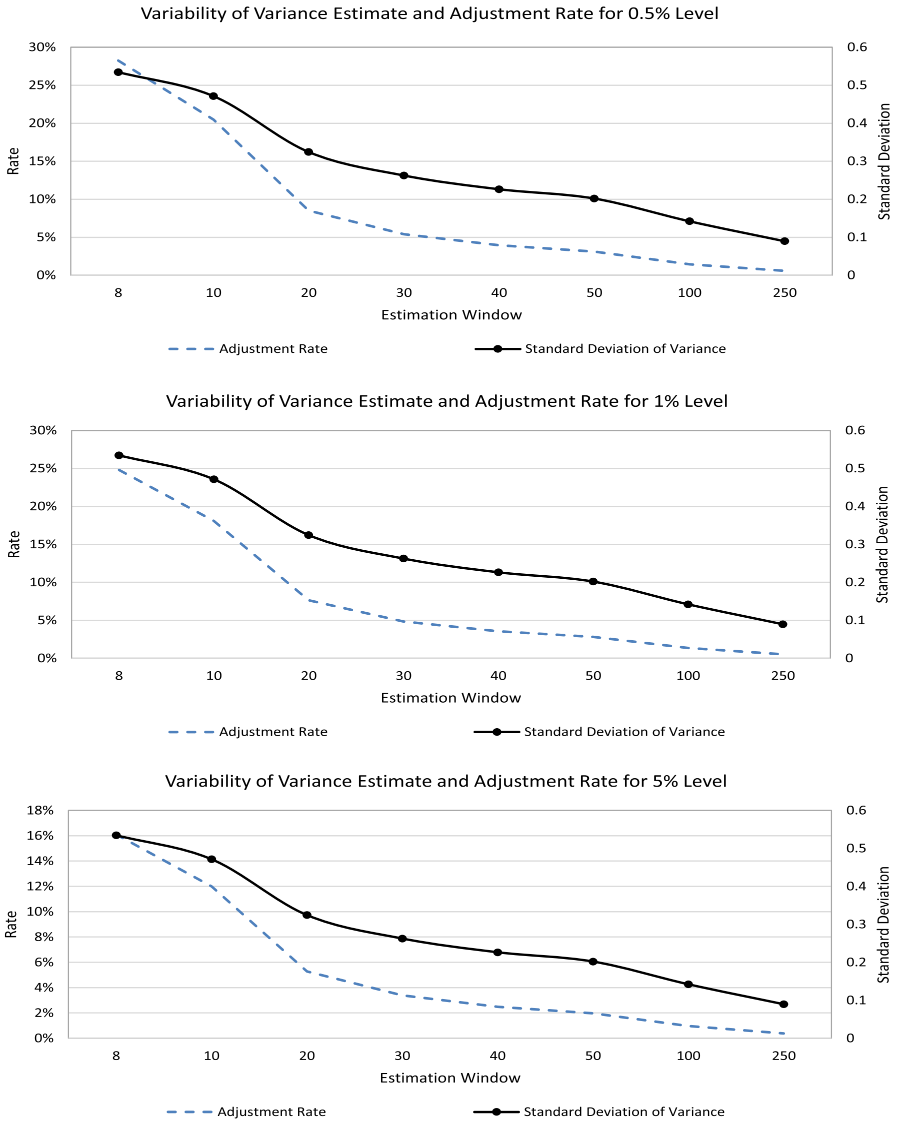


Fig. 2 Estimation window, variability of the variance estimate and adjustment rate

$$H_{0uc} \quad : \quad I_t \sim i.i.d.\ \text{Bernoulli}\,(\alpha)$$
against
$$H_{1uc} \quad : \quad I_t \sim i.i.d.\ \text{Bernoulli}\,(\pi)\,, \pi \neq \alpha$$

The null hypothesis H_0 states that the violation rate π is not different from the expected violation rate α, while the alternative states that both rates are different. The likelihood function is analytically derived, and the likelihood ratio test can be easily performed. Denote $T_1 = \sum_{t=1}^{T} I_t$ and $T_0 = \sum_{t=1}^{T} (1 - I_t) = T - T_1$.

The (log)-likelihood ratio statistic is given by

$$LR_{uc} = -2\ln[(1-\alpha)^{T_0}\alpha^{T_1} \big/ ((1 - T_1/T)^{T_0}(T_1/T)^{T_1})] \rightsquigarrow \chi_1^2$$

3.2 Conditional coverage

Following Christoffersen (1998), the independence assumption can also be tested along with the probability of violation accuracy.[9] Therefore, the alternative hypothesis should include the time dependence between I_t and I_{t+h}. We test

$$H_{0cc} : I_t \rightsquigarrow i.i.d. \text{ Bernoulli}(\alpha)$$

against

$$H_{1cc} : I_t \rightsquigarrow \text{Markov}(\Pi_1)\text{, with } \Pi_1 = \begin{bmatrix} \pi_{00} & \pi_{01} \\ \pi_{10} & \pi_{11} \end{bmatrix}$$

In contrast to the unconditional coverage, the null hypothesis H_{0cc} involves more restrictions in the parameters as the alternative includes more parameters. Denote by T_{ij} the number of observations of I_t with an i value followed by a j value. Formally, $T_{11} = \sum_{t=1}^{T-1} I_t I_{t+1}$, $T_{10} = \sum_{t=1}^{T-1} I_t (1 - I_{t+1})$, $T_{01} = \sum_{t=1}^{T-1} (1 - I_t) I_{t+1}$ and $T_{00} = \sum_{t=1}^{T-1} (1 - I_t)(1 - I_{t+1})$

The likelihood ratio for this hypothesis is given by

$$LR_{cc} = \begin{cases} -2\ln((1-\alpha)^{T_0}\alpha^{T_1} \big/ (1-\widehat{\pi}_{01})^{T_{00}}\widehat{\pi}_{01}^{T_{01}}(1-\widehat{\pi}_{11})^{T_{10}}\widehat{\pi}_{11}^{T_{11}}), \text{ if } \widehat{\pi}_{11} \neq 0 \\ \\ -2\ln((1-\alpha)^{T_0}\alpha^{T_1} \big/ (1-\widehat{\pi}_{01})^{T_{00}}\widehat{\pi}_{01}^{T_{01}}), \text{ if } \widehat{\pi}_{11} = 0 \end{cases}$$

where $\widehat{\pi}_{ij} = T_{ij} \big/ (T_{i0} + T_{i1})$ is the estimated probability that a j follows an i. The asymptotic distribution of LR_{cc} is a χ_2^2.

4 Adjusting for the violation rate bias

We propose here some adjustments alternatively on the level of the VaR or the expected level of violation rate. The overviolation comes from the randomness of σ_t, which leads to the following inequality

$$E\left[\Phi\left(-\frac{\widehat{\sigma}_t}{\sigma} z_\alpha\right)\right] > \alpha$$

To overcome this inequality, we can scale up σ_t to $\sigma_t^* = (1+\lambda)\,\sigma_t$ such that

$$E\left[\Phi\left(-\frac{(1+\lambda)\,\widehat{\sigma}_t}{\sigma} z_\alpha\right)\right] = \alpha \tag{4.1}$$

[9] Tests for multilevel can be found in Leccadito et al. (2014) and related papers

This is equivalent to use

$$VaR^*_\alpha (X_t) = (1 + \lambda)\, VaR_\alpha (X_t)$$

where $VaR_\alpha (X_t) = \sigma_t z_\alpha$. The problem is how to get λ. The solution to Equation (4.1) involves a double integration that is not readily solvable. When an analytical solution is intractable, alternative approaches such as Monte Carlo or quadrature techniques can be used to approximate the solution.

4.1 Adjusting the probability

Denoting by α^*, the theoretical expected rate of violation (i.e., $\alpha^* = E\,[\Pr(I_t = 1)]$), it follows that $E\,[I_t] = E\left[E\left(I_t \left|\widehat{\theta}_{t-1}\right.\right)\right] = E\,[\Pr(I_t = 1)] = \alpha^*.$, where $\widehat{\theta}_{t-1}$ represents the estimate of parameters used for the VaR forecast given available information. Since $I_t^2 = I_t$, we have that $\sigma^2\,[I_t] = \alpha^*\,(1 - \alpha^*)$ and the central limit theorem leads to

$$P^*_T = \frac{\sqrt{T}\left(\frac{1}{T}\sum_{t=1}^{T} I_t - \alpha^*\right)}{\sqrt{\alpha^*\,(1-\alpha^*)}} \xrightarrow{d} N\,(0, 1)$$

Therefore, testing whether the model is well specified implies testing the hypothesis

$$H_{0,\alpha^*} : E\,[I_t] = \alpha^*$$

It is worth noticing that although I_t is a binomial variable, the "conditional" probability of taking value 1 is not constant due to dynamic estimation of the VaR and the estimation risk. In other words, $\Pr(I_t = 1) = \alpha_t$ changes over time in our context. That is why both unconditional and conditional coverage backtestings described above cannot be used for the adjustment testing here even if replacing α with α^*. We should, however, notice that while both unconditional and conditional coverage backtestings are finite sample tests, the adjusting test procedure above requires a large sample.

In order to make a direct comparison with the unconditional coverage backtesting statistic, we can notice that the distribution of P_T^{*2} is a χ^2_1.

4.2 Adjusting the VaR estimate

Alternatively, we can correct the VaR estimate to make sure that the actual rate of violation is equal to α, the expected rate of violation. The adjustment should be made by introducing a scale factor λ^* as follows

$$VaR^*_\alpha = \left(1 + \lambda^*\right) VaR_\alpha (X_t)$$

with $I_t^* = I\left(X_t \leq -VaR_\alpha^*\right)$, and $E\left[I_t^*\right] = \alpha$. Therefore,

$$Q_T^* = \frac{\sqrt{T}\left(\frac{1}{T}\sum_T^{t=1} I_t^* - \alpha\right)}{\sqrt{\alpha\left(1-\alpha\right)}} \xrightarrow{d} N\left(0, 1\right)$$

In this case, testing whether the model is well specified implies testing the hypothesis

$$H_{0,\alpha} : E\left[I_t^*\right] = \alpha$$

Notice for the sake of comparison with previous statistic that Q_T^{*2} follows a χ_1^2. This second adjustment is more practical as it directly provides the level at which we should scale up the VaR forecasts to equate the actual violation rate with α, the expected rate.

4.3 Relating the adjustment to Basel II capital requirement

The Basel committee recommends a capital requirement for market risk to be $k \times VaR$ in excess of the specific risk of the financial institution. [10] The minimum value of k is 3, and after the backtesting, an adjustment is made depending on the result.[11] Assuming that the inaccuracy comes from estimation risk, the adjustment could be based on the process we described above. Therefore, the optimal scale factor would be determined by the value of the adjustment, λ^*. In the case of overviolation, the multiplier k could be replaced by $3(1 + \lambda^*)$ instead.

Table 1 presents the scaling factor used by regulators. The scaling is supposed to adjust the VaR to an acceptable range. This regulators scaling does not take into account the estimation error related to the framework used to estimate the VaR. Our suggestion is model-based and the scaling is endogenous with the goal of adjusting the VaR to a level that will be consistent with the expected violation rate.

In practice, the regulators require the use of 10-day VaR horizon for traded instruments. Therefore, a conversion of the VaR to a 10-day horizon is necessary. While we use the 1-day VaR in our application of Fama–French Portfolios' returns, the exercise can also be carried out in a similar way, with the only difference that the returns used are for a 10-day horizon. Using the $\sqrt{10}$ multiplied by the daily VaR can lead to an acceptable approximation, but it will not take into account the direct adjustment of the 10 days in the perspective of backtesting violation rate correction.

[10] In the 1996 Amendment, the capital requirement is given by the formula $\max(VaR_{t-t}, k \times VaR_{avg}) + SRC$, where k is a multiplicative factor, and SRC is a specific risk charge. VaR_{t-t} is the previous day's value-at-risk and VaR_{avg} is the average value-at-risk over the last 60 days (see Hull 2012).

[11] Table 1 provides the recommended scale factor by the Basle Committee on Banking Supervision for internal models.

4.4 Computing the adjustment

Using the quadrature techniques of Tauchen and Hussey (1991), we can obtain a very accurate approximation of numerical solution to Equation (4.1).[12] An N-point quadrature rule for integration of functions $g(x)$, $x \in \mathbb{R}$ against density f is a set of N points $\{x_k\}$ and corresponding N weights $\{f_k\}$ such that

$$\int g(x)\, f(x)\, dx \approx \sum_{k=1}^{N} g(x_k)\, f_k$$

where the symbol "≈" means approximation with convergence of the summation to the exact integral in the left-hand side of the equation. For numerical calculus, the number of points, their choice and the corresponding weights are very important for the accuracy of the method. A natural approach is to choose different quantiles (e.g., percentiles) and assign weights proportional to the value of the density at each corresponding points. Another approach is the classical N-point Gauss quadrature rule, which uses the method of moments to choose the points and the weights such that the N first moments of the discrete distribution match those of the continuous distribution. We combine both approaches as explained in "Appendix A".

The adjustment is based on solving a nonlinear equation; therefore, there is no guarantee for the side where the adjusted value will be, compared to the actual value.

The choice of the quadrature techniques here (instead of Monte Carlo) for computing the adjustment is driven by the fact that we are using Monte Carlo to analyze the overviolation and the effectiveness of the adjustment.

5 Monte Carlo study

To have a better understanding of the impact of estimation error on the violation rate, we perform simulation analysis. We simulate $m + T$ independent observations from a standard normal distribution. T is the sample of periods for the backtesting, while m represents the window of initial observations.

5.1 Setups

First, we generate T random observations X_τ, from the standard normal distribution $N(0, 1)$. We then estimate and update the volatility using different setups below:
(i) For each period, we simulate the standard error from a Chi-squared distribution with m degrees of freedom. This is actually the exact distribution of the sample estimate of the variance.

[12] The quadrature technique has been successfully used in different contexts by Balduzzi and Lynch (1999), Campbell and Viceira (1999) and Ang and Bekaert (2002). These authors find that the technique produces very accurate results even with few quadrature points.

(ii)-a. We perform the sample estimate of the standard deviation each time t using m more recent observations

$$\widehat{\sigma}_t = \sqrt{\frac{\sum_{\tau=t-m}^{t-1} \left(X_\tau - \overline{X}\right)^2}{m-1}}$$

This is the base case. It is a rolling window estimation using recent observation, and the theoretical adjustment factor can be derived as the distribution of the estimator is known.

(ii)-b. We apply the adjustment factor to the VaR estimate in ii)-a. This will allow us to assess the effectiveness of the adjustment in a well-designed setup.

(iii) The exponential weighted moving average (EWMA) is used as the updating framework for the standard deviation

$$\widehat{\sigma}_t^2 = 0.94\widehat{\sigma}_{t-1}^2 + 0.06X_{t-1}^2$$

While the EWMA updating puts the largest weight on the most recent return, the parameters are set regardless of the historical data. This is a practical advantage as there is no need for estimation and re-estimation at each period. However, the shortcoming of that model is the fact that one cannot statistically justify the parameter value used. The GARCH does not have this problem as the values of the parameters are estimated and can be re-estimated as data are updated.

(iv) The GARCH (1,1) is used as the updating framework for the standard deviation

$$\widehat{\sigma}_t^2 = \omega + \beta\widehat{\sigma}_{t-1}^2 + \alpha X_{t-1}^2$$

Note that in this case of the GARCH model, parameters have to be estimated using the maximum likelihood method. A distribution is therefore necessary for the innovations. In this Monte Carlo study, we use the normal distribution for consistency.

For the EWMA and GARCH setups, the initial value of the volatility is estimated using the first m observations. For the GARCH specification, the model is estimated once using all simulated data.[13]

For each probability level (0.5%, 1% and 5%), we simulate 1,000 runs of 10,000 observations each and use them to compute the violation rate for different windows and setups. Main results are presented in the section below.

5.2 Monte Carlo results

The increase in the estimation window reduces the estimation error. In fact, the standard deviation of the estimate of σ_t^2 in this case is $\sqrt{\frac{2}{m-1}}$. The derived adjustment factor

[13] Since this is a simulation exercise, there is no need to use an estimation sample for the GARCH parameters estimation, since all data are generated using the same DGP. For the empirical test, however, it is important to perform an out-of-sample exercise, by dividing the sample into two parts and using the initial subsample for estimation.

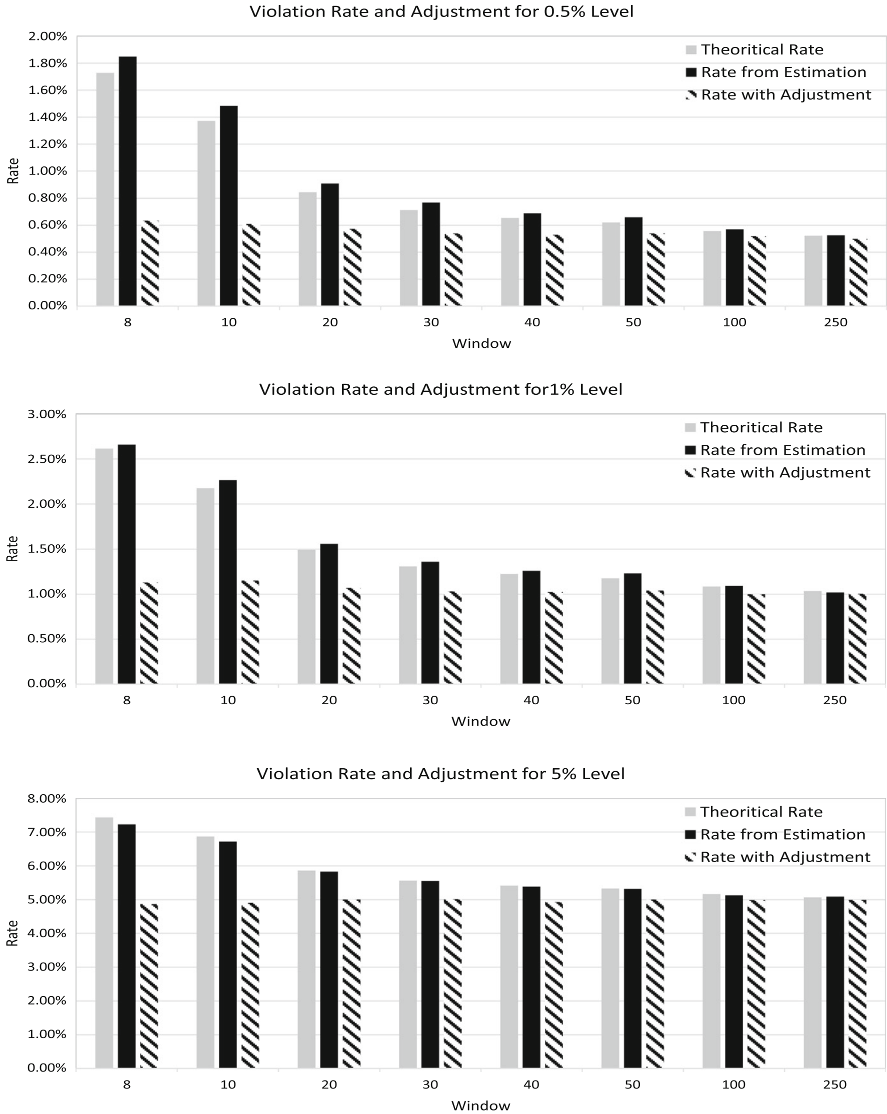


Fig. 3 Violation rate adjustment from the Monte Carlo study

also decreases when the window becomes larger (see Fig. 2). As the estimation error decreases, the discrepancy between actual and expected rates of violation decreases, and the adjustment needed for correction is smaller. For the 1% level, the adjustment rate moves from about 25% for an estimation window of eight observations to 0.5% for a window of 250 observations. The adjustment rate also decreases when the level of expected violation rate increases, moving from about 28% for a 0.5% level to about 16% for 5% level when we look at the eight-point estimation window.

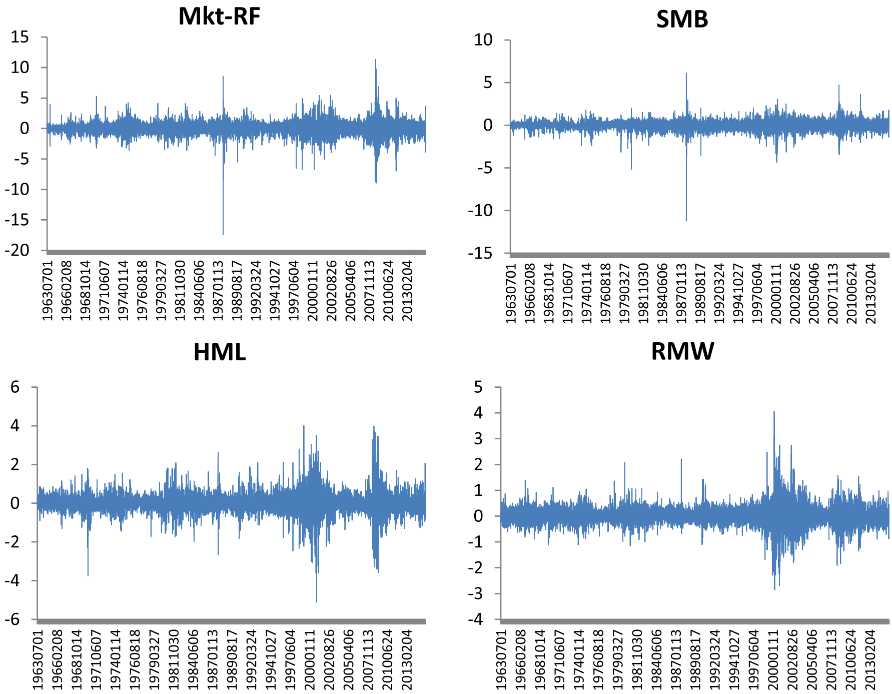


Fig. 4 Four Fama and French portfolio return series from July 1, 1963, to August 31, 2015

As we can see in Fig. 3, simulation analysis shows that the overviolation is larger as the window used for variance estimate is smaller. This is consistent with the fact that small window leads to a larger variation in the distribution of the variance. The theoretical expected violation rate is close to the rate obtained using simulated variance. This result shows the consistence between expected and actual overviolations.

Tables 4, 5 and 6 present Monte Carlo results for 0.5%, 1%, and 5% expected violation rates, respectively. The overviolation is more severe for 0.5% level (Table 4) moderate for 1% level (Table 5) and less severe for 5% level (Table 6). EWMA and GARCH models perform relatively better than the rolling window estimate of the variance for small windows, but underperform the latter for larger windows. We can notice that both EWMA and GARCH display a systematic (although less severe) overviolation for all three levels (0.5, 1% and 5%). The GARCH seems to perform relatively better than the EWMA. This can be explained by the fact that GARCH parameters are estimated and should display a better fit than the EWMA for which the updating parameter is set ad hoc. The adjustment brings the violation rate closer to the expected rate at all three levels and for almost any window length.

We can notice in this simulation exercise that the EWMA and the GARCH are good benchmarks. Although they also display overviolation rates, the issue is less severe and in most cases in line with the Basel Committee rules. Our main focus is the MA estimation which is often used with limited data. Our results suggest that the

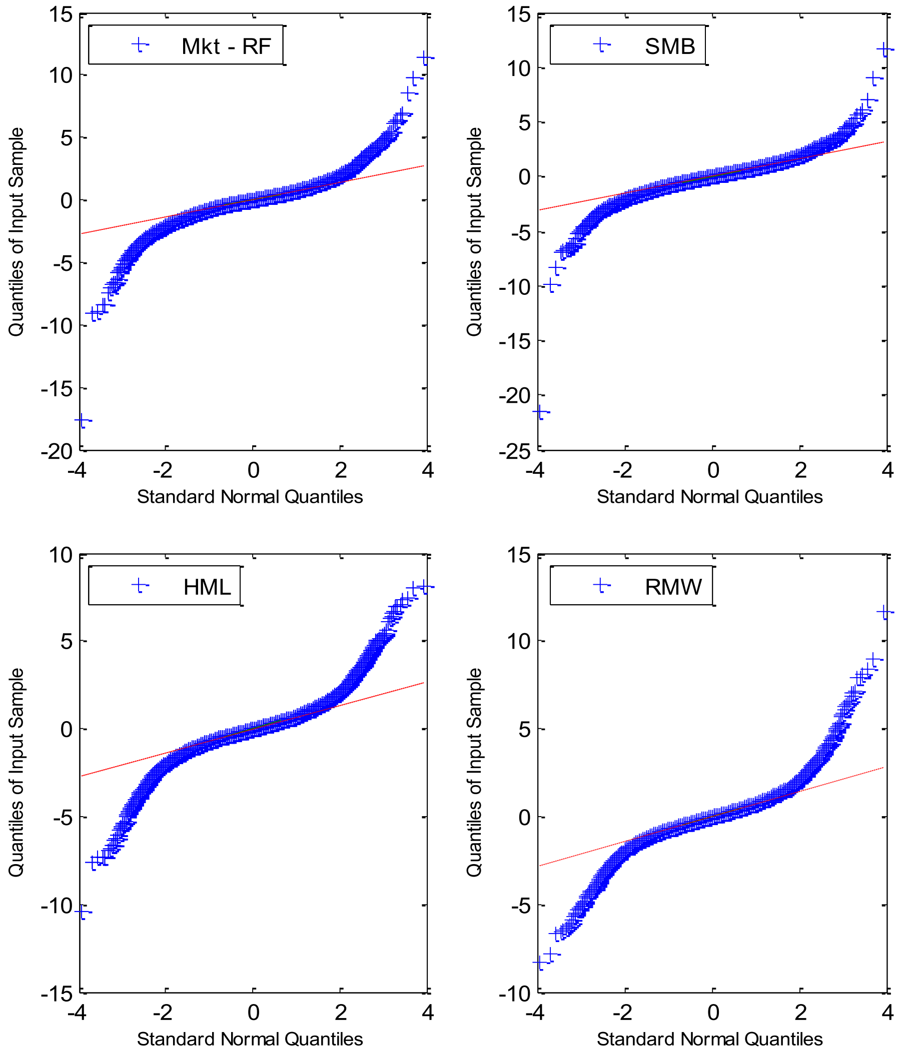


Fig. 5 QQ plot of (standardized) sample data versus standard normal distribution

overviolation rate could lead to a scaling factor by the regulators. The adjustment in most cases will bring the rate to the acceptable range.

6 Application to Fama–French portfolio returns

While the Monte Carlo study clearly shows the overviolation issue and the effectiveness of the proposed adjustment, the ultimate goal is to analyze this issue with the actual data. In the Monte Carlo study, the data are generated and behave as expected. The true distribution of actual data is not known and varies from one series to another.

The challenge is to investigate whether the issue is present in real data, to assess to which extent the adjustment is effective and when additional improvements are necessary to correct the issue. For the assessment tests, we use the conditional coverage (and unconditional coverage when the conditional coverage cannot be computed) for EWMA and GARCH because there is no adjustment. We do not use either conditional or unconditional coverage for both MA and adjusted MA. In fact, in the specific case of MA and adjusted MA, we used the two simple tests developed in Sect. 4: one based on the adjusted coverage (Sect. 4.1) and the other one based on the adjusted VaR (Sect. 4.2). Given the fact that these adjustments are computed in the MA framework, we could not use these tests for EWMA and GARCH models. That is why we use conditional or unconditional coverage tests. The test based on the adjusted coverage (Sect. 4.1) is used for the MA without adjustment, while the test based on the adjusted VaR (Sect. 4.2) is used for adjusted MA.

6.1 Data

We use four Fama–French portfolio returns: the market excess return (Mkt-RF), the size portfolio return (SMB), the book-to-market portfolio return (HML) and the momentum portfolio return (RMW). All data are daily from July 1, 1963, to August 31, 2015, for a total of 13,132 observations. Figure 4 provides a visual presentation of data dynamics.

Descriptive statistics are presented in Table 2. We notice that the kurtosis are larger than 3 for all series. This is an indication of fat tail in the data as we can see in the QQ plot in Fig. 5. The quantiles of sample data tend to diverge from the quantiles of standard normal distribution. To confirm that, we perform two tests: the Jarque–Bera test which is a moment-based test and the Kolmogorov–Smirnov test which is an empirical distribution-based test. Both tests strongly reject the normality of data (see results in Table 3).

6.2 Empirical results

All four portfolios provide consistent, but also specific results. As already observed in the Monte Carlo study, the overviolation issue is present when using actual data. The pattern is very similar as the issue is more severe for smaller windows.

Starting with the normal distribution, the presence of fat tail in the data does not allow the adjustment to appropriately correct the overviolation. For 0.05% VaR level (Table 7), even after applying the adjustment, the violation rate is still significantly above the expected rate and most often more than twice. The same issue is present for the 1% VaR level (Table 8) although less severe. At these more prudential levels, the EWMA and the GARCH perform relatively better.[14] This is clearly due to the ability of these models to capture unconditional tail in the data even under the normality assumption of the innovations. At a less prudential level (5% VaR in Table 9), the adjustment is more effective under the normal distribution. This suggests that normality is appropriate at some level, but fails to capture the data distribution behavior when we move deeply in the tails.

Replacing the normal distribution with the empirical distribution (Table 10) performs much better for more prudential levels.[15] At 0.5% VaR level, the empirical distribution improves the forecasts. Overviolation is more reasonable and the adjustment mostly corrects the issue, resulting in risk estimates which are mostly conservative. For 1% VaR level, the performance is also satisfactory. Results are less conservative at 1% VaR than for 0.5% VaR, but mostly accurate. The adjustment coupled with the empirical distribution significantly improves the risk forecasts for these prudential levels. Results are, however, less accurate for 5% VaR. At that level, the normal distribution works better. One explanation can be that, while data exhibit fat tails, they also have more concentration around the mean than the normal distribution does.

We did not apply the empirical distribution to the EWMA and the GARCH as the innovations for the likelihood estimation need a parametric distribution. The estimation of the GARCH model here assumes the normal distribution for innovations. Therefore, using the empirical distribution for VaR estimate will introduce some inconsistency.

Another important result come from the test using the theoretically derived probability (see p value in the brackets). Results from those tests follow the same pattern as the test based on adjustment. When the distribution is appropriate at least for the level in consideration, both adjustments perform relatively well.

Finally, we split our out-of-sample period in crisis and noncrisis (low-volatility) periods and find that the adjustment works much better for the empirical distribution

[14] Since our sample is from July 1, 1963, to August 31, 2015, we use the testing sample starting from January 20, 1976. We use data from July 1, 1963, to January 20, 1976, to estimate the GARCH model parameters. We perform some re-estimation check and realize that parameter tends to be very stable, so we do not feel the need to re-estimate at every time t.

[15] Empirical distribution is simply the appropriate quantile using past available data that are standardized (i.e., zero mean, standard deviation of one since the empirical distribution is standardized by taking the difference of each data value from the sample mean and dividing it by the sample standard deviation).

in each state separately than putting together (see Table 12). This may be due to the fact that distributions are different across different states of the economy.[16]

Although it is not our focus, the splitting will also help reduce the misspecification problem by allowing more flexibility in the distribution. In fact, splitting the sample can be seen as using a mixture of distribution or even a regime-switching dynamic model.

7 Conclusion

We investigate the effect of estimation error on the value-at-risk forecast. With a simple illustration, we show that an unbiased estimator of VaR can lead to overviolation of risk threshold and thus an underestimation of the risk. This overviolation is more severe when the estimation error is large. We show that this result is valid for normal distribution as well as for more general distribution. Some adjustments are proposed to account for this issue.

A Monte Carlo study illustrates the overviolation pattern and the effectiveness of the proposed adjustments. With a rolling window estimation of the variance used for VaR forecast, the overviolation is larger as the window is smaller. The adjustment scale factor also increases for more prudential levels of VaR estimates.

Applications to actual data highlight the need for further accounting for tail behavior. Under the normal distribution, the adjustment performs relatively well for a less prudential level (5% VaR), but fails for more prudential levels (1% or 0.5%VaR). The empirical distribution improves risk forecasts for more prudential levels.

While we shed a light on the overviolation and provide some attempts to address the issue, analyzing this issue in a more general framework involving multivariate distribution is an avenue that needs to be explored in further research.

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Compliance with Ethical Standards

Conflict of Interest The first author (Georges Tsafack) declares that he has no conflict of interest. The second author (James Cataldo) declares that he has no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

[16] The adjustment is related to the change in the estimation risk. And by splitting the sample in crisis versus noncrisis, the estimation can increase, leading to opposite direction adjustment in some instances.

Appendix A : Numerical solution using quadrature techniques

To be able to derive both the expected rate of violation α^* and the adjustment parameter λ^*, we employ quadrature techniques to discretize the distribution of σ_t. We first chose different quantiles $\{x_k\}$ $k = 1, \ldots N$. Then, we assign weights proportional to the value of the density of the true density $f(x_k)$ at each corresponding point such that the sum of weights is one. This provides us the starting point $\{w_{0k}\}$ for weights. To further improve accuracy, we create a mixture of this initial point with discrete equi-weight distribution as follows

$$f_k = (1-\varepsilon)\, w_{0k} + \varepsilon\,(1/N)$$

We then use the classical N-point Gauss quadrature rule to choose the weights such that the N first moments of the discrete distribution match those of the continuous distribution. Numerically, it is difficult to have the exact matching. We should minimize the sum of squared difference to obtain ε and the final weights. Alternatively, if the minimization does not provide appropriate results (due to corner solutions), an numerically appropriate value of ε can be set (Tables 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12).

Appendix B: Proof of Proposition 1

The rate of violation $r_\alpha = \frac{1}{T}\sum_{T}^{t=1} I_{\alpha,t}$; thus due to the stationarity, the expected rate of violation $E(r_\alpha) = E\left(I_{\alpha,t}\right)$. And by the law of iterate expectation, we have that

$$\begin{aligned}
E\left(I_{\alpha,t}\right) &= E\left[E\left[I_{\alpha,t}\,|\sigma_t\right]\right] \\
&= E\left[E\left[I\left(-X_t > \widehat{VaR}_\alpha(X_t)\right)|\sigma_t\right]\right] \\
&= E\left[E\left[I\left(X_t < -\widehat{\sigma}_t z_\alpha\right)|\sigma_t\right]\right] \\
&= E\left[\Pr\left[(X_t < -\sigma_t z_\alpha)\right]\right]
\end{aligned}$$

Since X_t follows a normal distribution $N\left(0,\sigma^2\right)$, the standardized variable $Z_t = X_t/\sigma$ follows the standard distribution $N(0,1)$.

Therefore,

$$\begin{aligned}
E\left[\Pr\left[(X_t < -\sigma_t z_\alpha)\right]\right] &= E\left[\Pr\left[(\sigma Z_t < -\sigma_t z_\alpha)\right]\right] \\
&= E\left[\Pr\left[\left(Z_t < -\frac{\sigma_t}{\sigma} z_\alpha\right)\right]\right] \\
&= E\left[\Phi\left(-\frac{\sigma_t}{\sigma} z_\alpha\right)\right]
\end{aligned}$$

Using the fact that Φ is a convex function (in the area of negative values), we have that

$$E\left[\Phi\left(-\frac{\sigma_t}{\sigma} z_\alpha\right)\right] > \Phi\left(E\left[-\frac{\sigma_t}{\sigma} z_\alpha\right]\right)$$

Table 1 Basel Committee boundaries and scaling factors for internal models backtesting

Zone	Number of exceptions	Increase in scaling factor	Cumulative probability (%)
Green zone	0	0.00	8.11
	1	0.00	28.58
	2	0.00	54.32
	3	0.00	75.81
	4	0.00	89.22
Yellow zone	5	0.40	95.88
	6	0.50	98.63
	7	0.65	99.60
	8	0.75	99.89
	9	0.85	99.97
Red zone	10 or more	1.00	99.99

The table defines the green, yellow and red zones that supervisors will use to assess backtesting results in conjunction with the internal models approach to market risk capital requirements. The boundaries shown in the table are based on a sample of 250 observations. For other sample sizes, the yellow zone begins at the point where the cumulative probability equals or exceeds 95%, and the red zone begins at the point where the cumulative probability equals or exceeds 99.99%

The cumulative probability is simply the probability of obtaining a given number or fewer exceptions in a sample of 250 observations when the true coverage level is 99%. For example, the cumulative probability shown for four exceptions is the probability of obtaining between zero and four exceptions

Note that these cumulative probabilities and the type 1 error probabilities reported in Table 1 do not sum to one because the cumulative probability for a given number of exceptions includes the possibility of obtaining exactly that number of exceptions, as does the type 1 error probability. Thus, the sum of these two probabilities exceeds one by the amount of the probability of obtaining exactly that number of exceptions

Source: This table comes from the Basel Committee on Banking Supervision (1996b) “Supervisory Framework for the Use of Backtesting in Conjunction with the Internal Models Approach to Market Risk Capital Requirements Bank for International Settlements” Policy Document

Table 2 Descriptive statistics

	Mkt-RF	SMB	HML	RMW
Mean	0.0238	0.0091	0.0170	0.0126
Standard error	0.0086	0.0045	0.0043	0.0030
Median	0.0500	0.0300	0.0100	0.0100
Standard deviation	0.9911	0.5201	0.4946	0.3474
Sample variance	0.9823	0.2705	0.2447	0.1207
Kurtosis	15.8624	22.4895	8.7904	9.5402
Skewness	− 0.5102	− 1.0081	0.0845	0.4048
Minimum	− 17.4400	− 11.1900	− 5.1200	− 2.8600
Maximum	11.3500	6.1000	4.0100	4.0600
Number of obs.	13,132	13,132	13,132	13,132

Mkt-RF is the market excess return, SMB the size portfolio return, HML the book-to-market portfolio return and RMW the momentum portfolio return

Table 3 Tests for normal distribution

	Mkt-RF	SMB	HML	RMW
Jarque–Bera statistic	138130	278750	42259	50117
(p value)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Kolmogorov–Smirnov statistic	0.4820	0.4908	0.4888	0.4918
(p value)	(0.0000)	(0.0000)	(0.0000)	(0.0000)

The null hypothesis in these tests states that the data are from a normal distribution. The Jarque–Bera test is an empirical moment based test, while Kolmogorov–Smirnov test is an empirical distribution-based test

$$= \Phi\left(-\frac{E\left[\sigma_t\right]}{\sigma} z_\alpha\right)$$

and using the concavity of square-root function $E\left[\sigma_t\right] = E\left[\sqrt{\sigma_t^2}\right] < \sqrt{E\left[\sigma_t^2\right]} = \sqrt{\sigma^2} = \sigma$. Therefore,

$$\begin{aligned}\Phi\left(-\frac{E\left[\sigma_t\right]}{\sigma} z_\alpha\right) &> \Phi\left(-\frac{\sigma}{\sigma} z_\alpha\right)\\ &= \Phi\left(-z_\alpha\right)\\ &= \alpha\end{aligned}$$

We can then conclude that for each t, $E\left(I_{\alpha,t}\right) > \alpha$ and the expected rate of violation $E\left(r_\alpha\right) > \alpha$. **Q.E.D**

Proof of Proposition 2

Since $\widehat{\upsilon}_\alpha$ is an unbiased estimate of the VaR. $E(\widehat{\upsilon}_\alpha)) = VaR_\alpha$

Table 4 Monte Carlo results for 0.5% expected violation rate (adjustment and rates are in percentage)

Window (m)		Adjustment	Theo. Rate	Simulate	MA	Adj. MA	EWMA	GARCH
8	Median	28.2601	1.7279	1.8250	1.8500	0.6350	0.7050	0.6200
	Mean	28.2601	1.7279	1.8295	1.8431	0.6411	0.7197	0.6246
	St_dev	0.0000	0.0000	0.1167	0.1366	0.0746	0.0862	0.0788
10	Median	20.4914	1.3719	1.4800	1.4850	0.6100	0.7200	0.6200
	Mean	20.4914	1.3719	1.4874	1.4838	0.6235	0.7154	0.6172
	St_dev	0.0000	0.0000	0.1027	0.0928	0.0709	0.0782	0.0762
20	Median	8.5016	0.8452	0.9300	0.9100	0.5750	0.7200	0.6100
	Mean	8.5016	0.8452	0.9238	0.9155	0.5768	0.7145	0.6153
	St_dev	0.0000	0.0000	0.0980	0.0907	0.0813	0.0733	0.0693
30	Median	5.4112	0.7132	0.7500	0.7700	0.5400	0.7250	0.6400
	Mean	5.4112	0.7132	0.7600	0.7630	0.5460	0.7244	0.6277
	St_dev	0.0000	0.0000	0.0845	0.0785	0.0750	0.0753	0.0774
40	Median	3.9635	0.6539	0.7050	0.6900	0.5300	0.7200	0.6100
	Mean	3.9635	0.6539	0.7003	0.6922	0.5358	0.7255	0.6249
	St_dev	0.0000	0.0000	0.0719	0.0802	0.0695	0.0770	0.0761
50	Median	3.1258	0.6204	0.6550	0.6600	0.5400	0.7400	0.6400
	Mean	3.1258	0.6204	0.6613	0.6586	0.5411	0.7358	0.6291
	St_dev	0.0000	0.0000	0.0826	0.0758	0.0737	0.0713	0.0689
100	Median	1.4472	0.5575	0.5700	0.5700	0.5200	0.7300	0.6300
	Mean	1.4472	0.5575	0.5703	0.5721	0.5175	0.7208	0.6256
	St_dev	0.0000	0.0000	0.0744	0.0717	0.0731	0.0835	0.0793

Table 4 continued

Window (m)		Adjustment	Theo. Rate	Simulate	MA	Adj. MA	EWMA	GARCH
250	Median	0.5856	0.5224	0.5250	0.5250	0.5000	0.7200	0.6250
	Mean	0.5856	0.5224	0.5310	0.5307	0.5050	0.7305	0.6301
	St_dev	0.0000	0.0000	0.0773	0.0708	0.0736	0.0778	0.0749

Adjustment is a scale factor to correct for overviolation, while the "Theo. Rate" is derived from the theoretical distribution of the violation indicator assuming normality for returns. "Simulate" are rates computed using the variance simulated from the Chi-square distribution. MA uses the variance estimate using rolling window. EWMA and GARCH are rates computed using these two respective models for variance update. Results are for 1000 runs of 10000 simulated returns

Table 5 Monte Carlo results for 1% expected violation rate (adjustment and rates are in percentage)

Window (m)		Adjustment	Theo. Rate	Simulate	MA	Adj. MA	EWMA	GARCH
8	Median	24.8087	2.6160	2.6300	2.6600	1.1300	1.3200	1.1800
	Mean	24.8087	2.6160	2.6315	2.6617	1.1242	1.3107	1.1712
	St_dev	0.0000	0.0000	0.1539	0.1488	0.1136	0.1033	0.1017
10	Median	18.0813	2.1772	2.2700	2.2650	1.1500	1.3200	1.1850
	Mean	18.0813	2.1772	2.2617	2.2643	1.1387	1.3188	1.1851
	St_dev	0.0000	0.0000	0.1464	0.1327	0.1025	0.1099	0.0965
20	Median	7.6294	1.4910	1.5350	1.5600	1.0700	1.3000	1.1500
	Mean	7.6294	1.4910	1.5438	1.5523	1.0705	1.2924	1.1533
	St_dev	0.0000	0.0000	0.1085	0.1104	0.0950	0.1011	0.0999
30	Median	4.8554	1.3086	1.3550	1.3600	1.0300	1.3200	1.1750
	Mean	4.8554	1.3086	1.3486	1.3636	1.0314	1.3149	1.1806
	St_dev	0.0000	0.0000	0.1232	0.0979	0.1048	0.0897	0.0916
40	Median	3.5585	1.2248	1.2550	1.2600	1.0250	1.3150	1.1700
	Mean	3.5585	1.2248	1.2531	1.2692	1.0329	1.3078	1.1755
	St_dev	0.0000	0.0000	0.1153	0.1024	0.0995	0.1047	0.0993
50	Median	2.8079	1.1767	1.2100	1.2300	1.0400	1.3300	1.2000
	Mean	2.8079	1.1767	1.2128	1.2255	1.0400	1.3207	1.1900
	St_dev	0.0000	0.0000	0.1204	0.1067	0.1097	0.1084	0.1036
100	Median	1.3662	1.0854	1.0750	1.0900	1.0000	1.3100	1.1700
	Mean	1.3662	1.0854	1.0862	1.0828	1.0032	1.2920	1.1554
	St_dev	0.0000	0.0000	0.1016	0.0875	0.1035	0.0957	0.0997

Table 5 continued

Window (m)		Adjustment	Theo. Rate	Simulate	MA	Adj. MA	EWMA	GARCH
250	Median	0.5301	1.0334	1.0300	1.0200	1.0050	1.2800	1.1500
	Mean	0.5301	1.0334	1.0324	1.0346	1.0024	1.2870	1.1603
	St_dev	0.0000	0.0000	0.1013	0.0976	0.1018	0.1049	0.1058

Adjustment is a scale factor to correct for overviolation, while the "Theo. Rate" is derived from the theoretical distribution of the violation indicator assuming normality for returns. "Simulate" are rates computed using the variance simulated from the Chi-square distribution. MA uses the variance estimate using rolling window. EWMA and GARCH are rates computed using these two respective models for variance update. Results are for 1000 runs of 10,000 simulated returns

Table 6 Monte Carlo results for 5% expected violation rate (adjustment and rates are in percentage)

Window (m)		Adjustment	Theo. Rate	Simulate	MA	Adj. MA	EWMA	GARCH
8	Median	16.1022	7.4372	7.1500	7.2300	4.8700	5.4850	5.2850
	Mean	16.1022	7.4372	7.1566	7.2042	4.8614	5.4669	5.2688
	St_dev	0.0000	0.0000	0.2693	0.2306	0.2446	0.1656	0.1805
10	Median	11.9954	6.8682	6.7000	6.7200	4.9050	5.4700	5.2550
	Mean	11.9954	6.8682	6.6942	6.7045	4.9015	5.4720	5.2726
	St_dev	0.0000	0.0000	0.2805	0.1973	0.2410	0.2132	0.2023
20	Median	5.2721	5.8614	5.8650	5.8300	5.0050	5.4800	5.2800
	Mean	5.2721	5.8614	5.8452	5.8389	4.9982	5.4703	5.2800
	St_dev	0.0000	0.0000	0.2306	0.2003	0.2114	0.2026	0.2007
30	Median	3.3795	5.5596	5.5350	5.5500	5.0150	5.4600	5.2650
	Mean	3.3795	5.5596	5.5505	5.5282	5.0028	5.4608	5.2797
	St_dev	0.0000	0.0000	0.2434	0.1909	0.2322	0.1899	0.1862
40	Median	2.4867	5.4144	5.3400	5.3850	4.9300	5.4600	5.2600
	Mean	2.4867	5.4144	5.3428	5.3883	4.9294	5.4683	5.2739
	St_dev	0.0000	0.0000	0.2035	0.1794	0.1988	0.1728	0.1874
50	Median	1.9670	5.3290	5.3200	5.3200	5.0050	5.4800	5.3100
	Mean	1.9670	5.3290	5.3127	5.3172	4.9950	5.4876	5.3022
	St_dev	0.0000	0.0000	0.2112	0.1899	0.2124	0.1937	0.1988
100	Median	0.9619	5.1621	5.1550	5.1300	4.9900	5.4650	5.2700
	Mean	0.9619	5.1621	5.1616	5.1321	5.0046	5.4590	5.2726
	St_dev	0.0000	0.0000	0.2050	0.1755	0.2031	0.1883	0.1868

Table 6 continued

Window (m)		Adjustment	Theo. Rate	Simulate	MA	Adj. MA	EWMA	GARCH
250	Median	0.3778	5.0642	5.0650	5.0900	4.9950	5.4800	5.3050
	Mean	0.3778	5.0642	5.0641	5.0734	5.0007	5.4833	5.2793
	St_dev	0.0000	0.0000	0.2153	0.1698	0.2146	0.1920	0.1892

Adjustment is a scale factor to correct for overviolation, while the "Theo. Rate" is derived from the theoretical distribution of the violation indicator assuming normality for returns. "Simulate" are rates computed using the variance simulated from the Chi-square distribution. MA uses the variance estimate using rolling window. EWMA and GARCH are rates computed using these two respective models for variance update. Results are for 1000 runs of 10,000 simulated returns

Table 7 Violation rates at 0.5% level, using the normal distribution (adjustment and rates are in percentage)

Window	Adjustment	Theo. Rate	Mkt-RF				SMB			
			MA	Adj. MA	EWMA	GARCH	MA	Adj. MA	EWMA	GARCH
8	28.2601	1.7279	2.5115	1.0506	1.2908	1.1007	2.4215	0.8205	0.9706	0.7805
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0011)
10	20.4914	1.3719	2.2418	1.1709	1.2910	1.1009	2.0216	0.9307	0.9708	0.7806
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0010)
20	8.5016	0.8452	1.5929	1.1120	1.2923	1.1120	1.3925	1.0018	0.9717	0.7914
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0007)
30	5.4112	0.7132	1.4240	1.0730	1.2936	1.1131	1.2736	1.0229	0.9727	0.7922
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0006)
40	3.9635	0.6539	1.4254	1.2648	1.2949	1.1142	1.1845	0.9837	0.9737	0.7830
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0010)
50	3.1258	0.6204	1.4570	1.2158	1.2962	1.1154	1.1455	1.0149	0.9747	0.7838
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0009)
100	1.4472	0.5575	1.3836	1.2826	1.3028	1.1210	1.1412	1.0301	0.9695	0.7776
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0013)
250	0.5856	0.5224	1.2818	1.2715	1.2818	1.1280	1.2818	1.2613	0.9844	0.7793
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0013)
			HML				RMW			
8	28.2601	1.7279	2.2313	0.8905	0.8305	0.7304	2.3514	0.9906	0.9506	0.8505
			(0.0000)	(0.0000)	(0.0000)	(0.0079)	(0.0000)	(0.0000)	0.0000	(0.0000)
10	20.4914	1.3719	1.8315	0.7706	0.8307	0.7306	1.8515	0.8907	0.9508	0.8507
			(0.0000)	(0.0001)	(0.0000)	(0.0079)	(0.0000)	(0.0000)	(0.0000)	(0.0000)

Table 7 continued

			HML				RMW			
20	8.5016	0.8452	1.1220	0.7413	0.8315	0.7313	1.3224	0.8716	0.9517	0.8515
			(0.0000)	(0.0006)	(0.0000)	(0.0077)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
30	5.4112	0.7132	0.9326	0.7120	0.8223	0.7320	1.1532	0.9426	0.9527	0.8524
			(0.0000)	(0.0027)	(0.0001)	(0.0075)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
40	3.9635	0.6539	0.8834	0.7227	0.8231	0.7328	1.1343	0.9536	0.9536	0.8532
			(0.0000)	(0.0016)	(0.0001)	(0.0074)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
50	3.1258	0.6204	0.7737	0.7034	0.8240	0.7335	1.0651	0.9747	0.9546	0.8541
			(0.0000)	(0.0040)	(0.0001)	(0.0072)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
100	1.4472	0.5575	0.7574	0.6968	0.8281	0.7372	1.0806	0.9897	0.9493	0.8483
			(0.0000)	(0.0055)	(0.0001)	(0.0023)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
250	0.5856	0.5224	0.8409	0.8203	0.8203	0.7383	1.0664	1.0664	0.9639	0.8614
			(0.0000)	(0.0000)	(0.0001)	(0.0023)	(0.0000)	(0.0000)	(0.0000)	(0.0000)

For each window and return series, the 0.5% VaR is estimated (i) using the rolling window estimation of the variance (MA); (ii) adjusting the VaR from MA to correct for estimation error (Adj. MA); (iii) using the EWMA; and (iv) using the GARCH model. The *p* values in parentheses are in (i) MA, for testing that the violation rate is equal to the theoretical rate using our proposed test; (ii) Adj. MA, for testing that the violation rate is equal to the 0.5% using our proposed test; and (iii) EWMA and GARCH for testing that the violation rate is equal to the 0.5% using the conditional coverage test and the unconditional coverage test when the former cannot be computed

Table 8 Violation rates at 1% level, using the normal distribution (adjustment and rates are in percentage)

Window	Adjustment	Theo. Rate	Mkt-RF				SMB			
			MA	Adj. MA	EWMA	GARCH	MA	Adj. MA	EWMA	GARCH
8	24.8087	2.6160	3.4821	1.7010	1.9712	1.7911	3.3920	1.4909	1.5709	1.5109
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
10	18.0813	2.1772	3.1325	1.7514	1.9716	1.7914	2.9023	1.5913	1.5713	1.5112
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
20	7.6294	1.4910	2.2841	1.7231	1.9736	1.7932	2.2440	1.6329	1.5728	1.5127
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
30	4.8554	1.3086	2.2463	1.7950	1.9755	1.7950	1.8753	1.5443	1.5744	1.5142
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
40	3.5585	1.2248	2.2184	1.9574	1.9775	1.7868	1.9072	1.6161	1.5760	1.5158
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
50	2.8079	1.1767	2.1302	1.8891	1.9795	1.7886	1.7986	1.6077	1.5776	1.5173
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
100	1.3662	1.0854	2.0804	1.9188	1.9693	1.7976	1.8885	1.7976	1.5754	1.5047
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
250	0.5301	1.0334	1.9586	1.8970	1.9586	1.7945	1.8970	1.8560	1.5792	1.5279
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	0.0000	(0.0000)	(0.0000)
			HML				RMW			
8	24.8087	2.6160	3.1919	1.4008	1.2508	1.1607	3.1819	1.4509	1.5309	1.3308
			(0.0000)	(0.0001)	(0.0044)	(0.0032)	(0.0000)	(0.0000)	(0.0000)	(0.0001)
10	18.0813	2.1772	2.6321	1.3811	1.2510	1.1609	2.6721	1.4912	1.5312	1.3311
			(0.0000)	(0.0001)	(0.0044)	(0.0032)	(0.0000)	(0.0000)	(0.0000)	(0.0001)

Table 8 continued

			HML				RMW			
20	7.6294	1.4910	1.7331	1.3625	1.2523	1.1521	2.0136	1.4526	1.5328	1.3324
			(0.0000)	(0.0003)	(0.0043)	(0.0033)	(0.0000)	(0.0000)	(0.0000)	(0.0001)
30	4.8554	1.3086	1.4440	1.2335	1.2435	1.1532	1.8652	1.4541	1.5343	1.3337
			(0.0000)	(0.0191)	(0.0050)	(0.0033)	(0.0000)	(0.0000)	(0.0000)	(0.0001)
40	3.5585	1.2248	1.4053	1.1845	1.2447	1.1544	1.7366	1.5158	1.5358	1.3351
			(0.0000)	(0.0642)	(0.0049)	(0.0033)	(0.0000)	(0.0000)	(0.0000)	(0.0001)
50	2.8079	1.1767	1.2761	1.1455	1.2460	1.1555	1.7283	1.5977	1.5374	1.3364
			(0.0008)	(0.1446)	(0.0048)	(0.0032)	(0.0000)	(0.0000)	(0.0000)	(0.0001)
100	1.3662	1.0854	1.3735	1.2523	1.2523	1.1614	1.6966	1.5552	1.5249	1.3432
			(0.0000)	(0.0116)	(0.0043)	(0.0030)	(0.0000)	(0.0000)	(0.0000)	(0.0001)
250	0.5301	1.0334	1.3023	1.2715	1.2510	1.1587	1.5279	1.4869	1.5279	1.3638
			(0.0000)	(0.0070)	(0.0043)	(0.0029)	(0.0000)	(0.0000)	(0.0000)	(0.0000)

For each window and return series, the 0.5% VaR is estimated i) using the rolling window estimation of the variance (MA); ii) adjusting the VaR from MA to correct for estimation error (Adj. MA); iii) using the EWMA; and iv) using the GARCH model. The p value in parentheses are in i) MA, for testing that the violation rate is equal to the theoretical rate using our proposed test; ii) Adj. MA, for testing that the violation rate is equal to the 0.5% using our proposed test; and iii) EWMA and GARCH for testing that the violation rate is equal to the 0.5% using the conditional coverage test and the unconditional coverage test when the former cannot be computed

Table 9 Violation rates at 5% level, using the normal distribution (adjustment and rates are in percentage)

Window	Adjustment	Theo. Rate	Mkt-RF				SMB			
			MA	Adj. MA	EWMA	GARCH	MA	Adj. MA	EWMA	GARCH
8	16.1022	7.4372	7.8747	5.7234	5.8935	5.8235	8.3650	5.9035	5.4132	5.3032
			(0.0000)	(0.0009)	(0.0001)	(0.0010)	(0.0000)	(0.0000)	(0.0000)	(0.0001)
10	11.9954	6.8682	7.3359	5.6345	5.8947	5.8247	7.8963	5.9648	5.4143	5.3042
			(0.0000)	(0.0036)	(0.0001)	(0.0010)	(0.0000)	(0.0000)	(0.0000)	(0.0001)
20	5.2721	5.8614	6.5518	5.8305	5.9006	5.8405	6.6319	5.8305	5.4198	5.3096
			(0.0000)	(0.0001)	(0.0001)	(0.0008)	(0.0000)	(0.0001)	(0.0000)	(0.0001)
30	3.3795	5.5596	6.2976	5.8163	5.9065	5.8564	6.3077	5.6959	5.4252	5.3049
			(0.0000)	(0.0002)	(0.0001)	(0.0006)	(0.0000)	(0.0014)	(0.0000)	(0.0002)
40	2.4867	5.4144	6.0530	5.7820	5.9125	5.8623	5.9727	5.5310	5.4306	5.3202
			(0.0000)	(0.0003)	(0.0001)	(0.0006)	(0.0000)	(0.0150)	(0.0000)	(0.0001)
50	1.9670	5.3290	6.0189	5.7978	5.9184	5.8581	5.8782	5.4863	5.4361	5.3256
			(0.0000)	(0.0003)	(0.0001)	(0.0006)	(0.0000)	(0.0260)	(0.0000)	(0.0001)
100	0.9619	5.1621	5.6251	5.4837	5.9079	5.8473	5.5342	5.3626	5.4534	5.3222
			(0.0000)	(0.0272)	(0.0001)	(0.0007)	(0.0000)	(0.0979)	(0.0000)	(0.0001)
250	0.3778	5.0642	5.5066	5.4758	5.9065	5.8860	5.4040	5.3527	5.4655	5.3835
			(0.0000)	(0.0311)	(0.0001)	(0.0004)	(0.0000)	(0.1100)	(0.0000)	(0.0000)
			HML				RMW			
8	16.1022	7.4372	7.8547	5.4933	5.2031	4.9730	8.2650	5.7835	5.3132	4.9230
			(0.0000)	(0.0237)	(0.0294)	(0.0181)	(0.0000)	(0.0003)	(0.0005)	(0.0006)
10	11.9954	6.8682	7.3759	5.4844	5.1841	4.9740	7.3859	5.6545	5.3143	4.9139
			(0.0000)	(0.0263)	(0.0284)	(0.0182)	(0.0000)	(0.0027)	(0.0005)	(0.0005)

Table 9 continued

			HML				RMW			
20	5.2721	5.8614	6.1411	5.1493	5.2094	4.9689	6.3815	5.5300	5.3196	4.9189
			(0.0000)	(0.4938)	(0.0016)	(0.0173)	(0.0000)	(0.0151)	(0.0005)	(0.0005)
30	3.3795	5.5596	5.5355	4.9840	5.1845	4.9639	5.7962	5.2948	5.3249	4.9238
			(0.0000)	(0.9414)	(0.0276)	(0.0164)	(0.0000)	(0.1767)	(0.0005)	(0.0005)
40	2.4867	5.4144	5.3102	4.9588	5.1696	4.9689	5.4909	5.2299	5.3303	4.9388
			(0.0009)	(0.8505)	(0.0263)	(0.0167)	(0.0000)	(0.2925)	(0.0005)	(0.0006)
50	1.9670	5.3290	5.1346	4.8432	5.1748	4.9739	5.4160	5.1547	5.3356	4.9337
			(0.0002)	(0.4731)	(0.0263)	(0.0170)	(0.0000)	(0.4788)	(0.0005)	(0.0006)
100	0.9619	5.1621	4.9687	4.8172	5.1808	4.9586	5.2616	5.1808	5.3121	4.9182
			(0.0000)	(0.4040)	(0.0247)	(0.0140)	(0.0000)	(0.4092)	(0.0004)	(0.0004)
250	0.3778	5.0642	4.8913	4.8503	5.1579	4.9938	5.1989	5.1887	5.3015	4.8913
			(0.0000)	(0.4975)	(0.0297)	(0.0236)	(0.0000)	(0.3926)	(0.0003)	(0.0005)

For each window and return series, the 5% VaR is estimated i) using the rolling window estimation of the variance (MA); ii) adjusting the VaR from MA to correct for estimation error (Adj. MA); iii) using the EWMA; and iv) using the GARCH model. The *p* values in parentheses are in i) MA, for testing that the violation rate is equal to the theoretical rate using our proposed test; ii) Adj. MA, for testing that the violation rate is equal to the 5% using our proposed test; and iii) EWMA and GARCH for testing that the violation rate is equal to the 5% using the conditional coverage test and the unconditional coverage test when the former cannot be computed

Table 10 Violation rates using the empirical distribution (adjustment and rates are in percentage)

Window	0.5%			1%			5%		
	Theo. Rate	MA	Adj. MA	Theo. Rate	MA	Adj. MA	Theo. Rate	MA	Adj. MA
Mkt-RF									
8	1.7279	0.7705	0.2401	2.6160	1.7511	0.8305	7.4372	8.3750	6.1337
		(0.0016)	(0.0002)		(0.0000)	(0.0886)		(0.0000)	(0.0000)
10	1.3719	0.5604	0.2902	2.1772	1.4912	0.7306	6.8682	7.8763	6.0649
		(0.0882)	(0.0030)		(0.0000)	(0.0068)		(0.0000)	(0.0000)
20	0.8452	0.3106	0.1903	1.4910	1.0218	0.7514	5.8614	7.0327	6.2913
		(0.0039)	(0.0000)		(0.0029)	(0.0125)		(0.0000)	(0.0000)
30	0.7132	0.3008	0.2106	1.3086	0.8724	0.7521	5.5596	6.7288	6.2475
		(0.0021)	(0.0000)		(0.0123)	(0.0128)		(0.0000)	(0.0000)
40	0.6539	0.3011	0.2409	1.2248	0.8332	0.7729	5.4144	6.4746	6.0932
		(0.0021)	(0.0002)		(0.0048)	(0.0227)		(0.0000)	(0.0000)
50	0.6204	0.2613	0.2311	1.1767	0.8039	0.6933	5.3290	6.4811	6.1596
		(0.0002)	(0.0001)		(0.0003)	(0.0021)		(0.0000)	(0.0000)
100	0.5575	0.2828	0.2828	1.0854	0.8382	0.7978	5.1621	6.0594	5.9887
		(0.0007)	(0.0022)		(0.0000)	(0.0432)		(0.0000)	(0.0000)
250	0.5224	0.3589	0.3384	1.0334	0.8511	0.8101	5.0642	5.9783	5.9270
		(0.0000)	(0.0237)		(0.0001)	(0.0595)		(0.0000)	(0.0000)
HML									
8	1.7279	1.5409	0.6304	2.6160	2.6516	1.1807	7.4372	7.7146	5.4132
		(0.0000)	(0.0646)		(0.0000)	(0.0694)		(0.0000)	(0.0580)
10	1.3719	1.2010	0.4704	2.1772	2.1117	1.1309	6.8682	7.2158	5.3943
		(0.0000)	(0.6746)		(0.0000)	(0.1885)		(0.0000)	(0.0705)
20	0.8452	0.6912	0.5510	1.4910	1.4226	1.0519	5.8614	6.0309	5.0791
		(0.0095)	(0.4701)		(0.0000)	(0.6023)		(0.0000)	(0.7168)

Table 10 continued

Window	0.5%			1%			5%		
	Theo. Rate	MA	Adj. MA	Theo. Rate	MA	Adj. MA	Theo. Rate	MA	Adj. MA
30	0.7132	0.5415	0.4412	1.3086	1.1833	0.9426	5.5596	5.4252	4.9138
		(0.4942)	(0.4054)		(0.0005)	(0.5648)		(0.0002)	(0.6927)
40	0.6539	0.5822	0.4316	1.2248	1.0942	0.9536	5.4144	5.2299	4.8585
		(0.0749)	(0.3334)		(0.0195)	(0.6418)		(0.0013)	(0.5169)
50	0.6204	0.5527	0.4622	1.1767	0.9948	0.8641	5.3290	5.0241	4.7629
		(0.0908)	(0.5931)		(0.0022)	(0.1732)		(0.0002)	(0.2777)
100	0.5575	0.4848	0.4646	1.0854	1.0402	0.9796	5.1621	4.8475	4.7364
		(0.0004)	(0.6170)		(0.0005)	(0.8383)		(0.0000)	(0.2288)
250	0.5224	0.6050	0.5640	1.0334	1.1177	1.0767	5.0642	4.8195	4.7888
		(0.0076)	(0.3703)		(0.0006)	(0.4465)		(0.0000)	(0.3385)
SMB									
8	1.7279	0.8605	0.2802	2.6160	2.2914	0.8605	7.4372	10.5563	7.7146
		(0.0000)	(0.0018)		(0.0000)	(0.1611)		(0.0000)	(0.0000)
10	1.3719	0.7806	0.3403	2.1772	1.8114	0.9307	6.8682	9.7678	7.8763
		(0.0000)	(0.0236)		(0.0000)	(0.4866)		(0.0000)	(0.0000)
20	0.8452	0.4408	0.2104	1.4910	1.3224	0.9517	5.8614	8.3550	7.6438
		(0.2953)	(0.0000)		(0.0028)	(0.6278)		(0.0000)	(0.0000)
30	0.7132	0.3209	0.1805	1.3086	1.1733	0.9426	5.5596	8.0726	7.5411
		(0.0066)	(0.0000)		(0.0123)	(0.5648)		(0.0000)	(0.0000)
40	0.6539	0.3413	0.2510	1.2248	1.0540	0.9235	5.4144	7.9602	7.5085
		(0.0160)	(0.0004)		(0.0835)	(0.4429)		(0.0000)	(0.0000)
50	0.6204	0.3818	0.2814	1.1767	1.0551	0.9646	5.3290	7.8175	7.4256
		(0.0728)	(0.0020)		(0.0192)	(0.7229)		(0.0000)	(0.0000)

Table 10 continued

Window	0.5%			1%			5%		
	Theo. Rate	MA	Adj. MA	Theo. Rate	MA	Adj. MA	Theo. Rate	MA	Adj. MA
100	0.5575	0.4040	0.3535	1.0854	1.0200	0.9695	5.1621	7.2713	7.1299
		(0.0140)	(0.0387)		(0.0000)	(0.7604)		(0.0000)	(0.0000)
250	0.5224	0.4307	0.4102	1.0334	1.2100	1.1895	5.0642	6.8909	6.8499
		(0.0018)	(0.2085)		(0.0000)	(0.0600)		(0.0000)	(0.0000)
RMW									
8	1.7279	1.5809	0.6204	2.6160	2.8417	1.2708	7.4372	9.6358	6.9942
		(0.0000)	(0.0880)		(0.0000)	(0.0065)		(0.0000)	(0.0000)
10	1.3719	1.2710	0.6005	2.1772	2.3619	1.2710	6.8682	8.9071	6.9255
		(0.0000)	(0.1544)		(0.0000)	(0.0065)		(0.0000)	(0.0000)
20	0.8452	0.7614	0.5810	1.4910	1.6630	1.2923	5.8614	7.8040	6.9124
		(0.0000)	(0.2510)		(0.0000)	(0.0033)		(0.0000)	(0.0000)
30	0.7132	0.7722	0.5415	1.3086	1.5343	1.2435	5.5596	7.3606	6.6586
		(0.0001)	(0.5567)		(0.0000)	(0.0145)		(0.0000)	(0.0000)
40	0.6539	0.6224	0.5320	1.2248	1.5158	1.2548	5.4144	6.8661	6.4445
		(0.0000)	(0.6505)		(0.0000)	(0.0106)		(0.0000)	(0.0000)
50	0.6204	0.6732	0.5326	1.1767	1.5273	1.2761	5.3290	6.7725	6.4510
		(0.0000)	(0.6452)		(0.0000)	(0.0056)		(0.0000)	(0.0000)
100	0.5575	0.6867	0.6564	1.0854	1.3735	1.3028	5.1621	6.3624	6.2715
		(0.0000)	(0.0273)		(0.0000)	(0.0025)		(0.0000)	(0.0000)
250	0.5224	0.6870	0.6768	1.0334	1.2715	1.2715	5.0642	6.3884	6.3269
		(0.0000)	(0.0133)		(0.0000)	(0.0070)		(0.0000)	(0.0000)

For each window and return series, the VaR is estimated i) using the rolling window estimation of the variance (MA) and ii) adjusting the VaR from MA to correct for estimation error (Adj. MA. The *p* values in parentheses are in i) MA, for testing that the violation rate is equal to the theoretical rate using our proposed test; ii) Adj. MA, for testing that the violation rate is equal to the corresponding expected violation rate level using our proposed test

Table 11 Violation rates using the normal distribution (in percentage): crisis versus noncrisis periods

Window	Crisis period						Non-crisis period					
	0.5%		1%		5%		0.5%		1%		5%	
	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA
Mkt-Rf												
8	2.6093	1.1283	3.2440	1.5515	8.8152	5.9238	2.5283	1.0431	3.5374	1.7347	7.7438	5.6803
	(0.0000)	(0.0008)	(0.0000)	(0.0369)	(0.0000)	(0.1104)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0034)
10	2.1186	1.2006	2.8249	1.9068	7.9802	5.7203	2.2794	1.1907	3.1867	1.7578	7.2465	5.6135
	(0.0000)	(0.0002)	(0.0000)	(0.0006)	(0.0000)	(0.2136)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0082)
20	1.9203	1.4936	2.5605	2.1337	7.1835	6.4011	1.5441	1.0559	2.2480	1.6576	6.4260	5.7107
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0004)	(0.0159)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0241)	(0.0022)
30	2.1490	1.5043	2.6504	2.3639	7.3782	6.9484	1.3185	1.0116	2.1823	1.7163	6.1150	5.6263
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0008)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0230)	(0.0070)
40	1.9481	1.8038	3.0303	2.5253	7.0707	6.8543	1.3427	1.1834	2.0938	1.8776	5.8716	5.5985
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0008)	(0.0015)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0003)	(0.0100)
50	2.0349	1.8895	2.7616	2.6163	6.9041	6.7587	1.3671	1.1164	2.0392	1.7886	5.8783	5.6391
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0021)	(0.0028)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0002)	(0.0060)
100	1.9608	1.9608	2.8658	2.5641	6.6365	6.4103	1.3061	1.1916	1.9592	1.8217	5.4652	5.3391
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0091)	(0.0185)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0493)	(0.1460)
250	2.2109	2.1259	3.4014	3.2313	7.9082	7.8231	1.1541	1.1541	1.7603	1.7137	5.1760	5.1527
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.4569)	(0.5164)
HML												
8	2.4683	1.6220	3.3145	1.9041	7.8279	5.6417	2.1995	0.7937	3.1746	1.3379	7.9365	5.5329
	(0.0000)	(0.0000)	(0.0000)	(0.0006)	(0.0000)	(0.2675)	(0.0000)	(0.0001)	(0.0000)	(0.0014)	(0.0000)	(0.0217)
10	2.0480	1.0593	2.8955	1.6949	7.4153	6.0028	1.8031	0.7371	2.5970	1.3268	7.4393	5.4548

Table 11 continued

Window	Crisis period						Non-crisis period					
	0.5%		1%		5%		0.5%		1%		5%	
	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA
	(0.0000)	(0.0028)	(0.0000)	(0.0086)	(0.0001)	(0.0834)	(0.0000)	(0.0016)	(0.0000)	(0.0020)	(0.0000)	(0.0501)
20	1.3514	1.1380	2.1337	1.7781	6.3300	5.6188	1.1013	0.6812	1.7030	1.3170	6.1762	5.1203
	(0.0002)	(0.0007)	(0.0002)	(0.0034)	(0.0277)	(0.2871)	(0.0086)	(0.0159)	(0.1006)	(0.0028)	(0.2085)	(0.6043)
30	1.0745	0.8596	1.6476	1.5043	5.5158	4.7994	0.9093	0.6933	1.3980	1.1821	5.5808	5.0693
	(0.0084)	(0.0568)	(0.0261)	(0.0583)	(0.3841)	(0.7310)	(0.0288)	(0.0101)	(0.4603)	(0.0861)	(0.9309)	(0.7654)
40	1.0823	0.9380	1.5873	1.4430	4.9062	4.7619	0.8534	0.6941	1.3655	1.1379	5.3482	4.9727
	(0.0078)	(0.0208)	(0.0429)	(0.0974)	(0.8723)	(0.6842)	(0.0000)	(0.0099)	(0.0011)	(0.1938)	(0.1384)	(0.9065)
50	1.0174	0.9448	1.5988	1.4535	4.6512	4.3605	0.7405	0.6721	1.2303	1.1050	5.1834	4.8872
	(0.0171)	(0.0193)	(0.0400)	(0.0909)	(0.5482)	(0.2764)	(0.0029)	(0.0222)	(0.0363)	(0.3226)	(0.4331)	(0.6278)
100	1.2066	1.1312	2.0362	1.6591	4.6757	4.4495	0.7104	0.6531	1.2832	1.2030	4.9954	4.8465
	(0.0020)	(0.0011)	(0.0009)	(0.0159)	(0.5840)	(0.3577)	(0.0088)	(0.0426)	(0.0108)	(0.0566)	(0.9843)	(0.5105)
250	1.8707	1.8707	2.2109	2.2109	5.4422	5.2721	0.6995	0.6761	1.1774	1.1425	4.8146	4.7913
	(0.0000)	(0.0000)	(0.0003)	(0.0000)	(0.4925)	(0.6685)	(0.0135)	(0.0207)	(0.1082)	(0.1848)	(0.4281)	(0.3752)

Table 11 continued

Window	Crisis period						Non-crisis period					
	0.5%		1%		5%		0.5%		1%		5%	
	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA
SMB												
8	2.3977	0.7052	3.1735	1.3399	8.4626	5.2891	2.4036	0.8277	3.4014	1.4966	8.3220	5.9977
	(0.0000)	(0.2732)	(0.0000)	(0.1983)	(0.0000)	(0.6174)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
10	1.9774	0.8475	2.8955	1.8362	7.6271	5.5791	2.0186	0.9413	2.8691	1.5536	7.9156	5.9991
	(0.0000)	(0.0638)	(0.0000)	(0.0016)	(0.0000)	(0.3174)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
20	1.4936	1.0669	2.4893	1.6358	6.6856	5.8321	1.3624	0.9877	2.1912	1.6122	6.5849	5.8015
	(0.0000)	(0.0026)	(0.0000)	(0.0166)	(0.0057)	(0.1522)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0038)	(0.0006)
30	1.2894	0.9312	1.7908	1.4327	6.5903	6.0888	1.2617	1.0230	1.8868	1.5572	6.2173	5.6035
	(0.0005)	(0.0224)	(0.0075)	(0.1042)	(0.0092)	(0.0620)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0071)	(0.0094)
40	1.0823	0.9380	1.5873	1.2987	6.0606	5.6277	1.1948	0.9900	1.9345	1.6500	5.8944	5.4620
	(0.0078)	(0.0208)	(0.0429)	(0.2637)	(0.0791)	(0.2836)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0002)	(0.0469)
50	1.0901	1.0901	1.7442	1.5262	5.5233	5.3052	1.1506	1.0025	1.8000	1.6177	5.8897	5.4796
	(0.0073)	(0.0019)	(0.0121)	(0.0498)	(0.3808)	(0.6034)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0002)	(0.0392)
100	1.2066	1.0558	2.2624	2.0362	5.8824	5.6561	1.1114	1.0082	1.8103	1.7415	5.4537	5.2933
	(0.0020)	(0.0041)	(0.0001)	(0.0001)	(0.1510)	(0.2730)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0551)	(0.2086)
250	2.8912	2.8912	3.5714	3.4864	7.3980	7.3980	1.0609	1.0375	1.6671	1.6321	5.1294	5.0711
	(0.0000)	(0.0000)	(0.0000)	0.0000	(0.0004)	(0.0002)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.5839)	(0.7625)
RWM												
8	2.6798	1.1283	3.5261	1.6220	8.8152	6.2059	2.2902	0.9524	3.1293	1.4172	8.1859	5.7483
	(0.0000)	(0.0008)	(0.0000)	(0.0186)	(0.0000)	(0.0372)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0000)	(0.0013)
10	2.3305	1.0593	3.1780	1.8362	7.5565	6.0028	1.7691	0.8619	2.6083	1.4289	7.3713	5.6135

Table 11 continued

Window	Crisis period						Non-crisis period					
	0.5%		1%		5%		0.5%		1%		5%	
	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA
	(0.0000)	(0.0028)	(0.0000)	(0.0016)	(0.0000)	(0.0834)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0000)	(0.0082)
20	1.7781	1.0669	2.4182	1.8492	6.3300	5.6188	1.2829	0.8288	1.9868	1.4192	6.4260	5.5518
	(0.0000)	(0.0026)	(0.0000)	(0.0014)	(0.0277)	(0.2871)	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.0241)	(0.0175)
30	1.4327	1.2178	2.2923	1.7192	5.7307	5.3009	1.1139	0.8979	1.8072	1.4208	5.8536	5.3308
	(0.0001)	(0.0001)	(0.0000)	(0.0069)	(0.2204)	(0.6060)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.2288)	(0.1546)
40	1.2987	1.0823	2.3810	1.9481	5.2670	5.1227	1.1152	0.9331	1.6614	1.4679	5.5530	5.2799
	(0.0005)	(0.0021)	(0.0000)	(0.0004)	(0.6511)	(0.8340)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0193)	(0.2286)
50	1.0901	1.0174	2.2529	2.1076	5.3052	5.1599	1.0595	0.9683	1.6632	1.5152	5.4682	5.1948
	(0.0073)	(0.0065)	(0.0001)	(0.0000)	(0.6068)	(0.7855)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0472)	(0.4023)
100	1.2821	1.2821	2.1870	1.8854	4.9774	4.8265	1.0426	0.9395	1.6155	1.5009	5.3048	5.2360
	(0.0008)	(0.0001)	(0.0002)	(0.0012)	(0.9698)	(0.7720)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.1956)	(0.3117)
250	2.0408	2.0408	2.4660	2.3810	6.2925	6.2925	0.9326	0.9326	1.3989	1.3640	5.0478	5.0361
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0502)	(0.0420)	(0.0000)	(0.0000)	(0.0005)	(0.0007)	(0.8393)	(0.8779)

The left panels represent tests over the crisis periods marked by the NBER since 1980 (the last 5 recessions). For each window and return series, the VaR is estimated i) using the rolling window estimation of the variance (MA) and ii) adjusting the VaR from MA to correct for estimation error (Adj. MA. The *p* values in parentheses are in i) MA, for testing that the violation rate is equal to the theoretical rate using our proposed test; ii) Adj. MA, for testing that the violation rate is equal to the corresponding expected violation rate level using our proposed test

Table 12 Violation rates using the empirical distribution (in percentage): crisis versus noncrisis periods

Window	Crisis period						Non-crisis period					
	0.5%		1%		5%		0.5%		1%		5%	
	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA
Mkt-Rf												
8	0.8463	0.2116	1.6925	0.9168	9.3089	6.5585	0.7710	0.2381	1.7914	0.8277	8.2313	6.0658
	(0.0927)	(0.1236)	(0.0171)	(0.7528)	(0.0000)	(0.0071)	(0.0008)	(0.0005)	(0.0000)	(0.1038)	(0.0000)	(0.0000)
10	0.5650	0.3531	1.6949	0.7768	8.5452	6.3559	0.5670	0.2835	1.4856	0.7371	7.7795	6.0104
	(0.7342)	(0.4332)	(0.0168)	(0.3987)	(0.0000)	(0.0192)	(0.3824)	(0.0039)	(0.0000)	(0.0131)	(0.0000)	(0.0000)
20	0.4267	0.2134	1.3514	0.9957	7.7525	6.9701	0.2838	0.1817	0.9764	0.7153	6.8915	6.1649
	(0.6895)	(0.1276)	(0.2088)	(0.9872)	(0.0000)	(0.0007)	(0.0000)	(0.0000)	(0.0001)	(0.0072)	(0.0000)	(0.0000)
30	0.3582	0.2865	1.2894	0.9312	8.0229	7.3782	0.2842	0.1932	0.8070	0.7161	6.5242	6.0582
	(0.4286)	(0.2581)	(0.2982)	(0.7962)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0074)	(0.0001)	(0.0000)
40	0.3608	0.3608	1.2987	1.0823	7.5758	7.0707	0.2845	0.2162	0.7624	0.7169	6.2927	5.9172
	(0.4392)	(0.4623)	(0.2853)	(0.7583)	(0.0000)	(0.0004)	(0.0018)	(0.0002)	(0.0195)	(0.0076)	(0.0000)	(0.0001)
50	0.3634	0.3634	1.1628	0.9448	7.4855	6.9767	0.2392	0.2051	0.7405	0.6494	6.3112	6.0264
	(0.4500)	(0.4724)	(0.5541)	(0.8369)	(0.0001)	(0.0008)	(0.0001)	(0.0001)	(0.0104)	(0.0010)	(0.0000)	(0.0000)
100	0.4525	0.4525	1.5837	1.5083	7.0890	7.0136	0.2521	0.2521	0.7218	0.6874	5.8891	5.8203
	(0.8031)	(0.8062)	(0.0490)	(0.0629)	(0.0010)	(0.0008)	(0.0003)	(0.0010)	(0.0060)	(0.0033)	(0.0002)	(0.0004)
250	0.9354	0.8503	1.6156	1.5306	8.9286	8.7585	0.2798	0.2681	0.7461	0.7111	5.5724	5.5374
	(0.0591)	(0.0885)	(0.0514)	(0.0674)	(0.0000)	(0.0000)	(0.0016)	(0.0023)	(0.0133)	(0.0072)	(0.0168)	(0.0224)
HML												
8	2.0451	1.1989	7.8279	5.5712	7.8279	5.5712	1.4739	0.5669	2.6304	1.1224	7.7778	5.4535
	(0.0000)	(0.0002)	(0.0000)	(0.3237)	(0.0000)	(0.3237)	(0.0000)	(0.3731)	(0.0000)	(0.2478)	(0.0000)	(0.0507)
10	1.4831	0.8475	7.2740	6.0028	7.2740	6.0028	1.1567	0.4309	2.0866	1.1000	7.2579	5.3527
	(0.0000)	(0.0638)	(0.0002)	(0.0834)	(0.0002)	(0.0834)	(0.0000)	(0.3579)	(0.0000)	(0.3452)	(0.0000)	(0.1286)

Table 12 continued

Window	Crisis period						Non-crisis period					
	0.5%		1%		5%		0.5%		1%		5%	
	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA
20	1.0669	1.0669	6.2589	5.5477	6.2589	5.5477	0.6358	0.4768	1.3738	1.0218	6.0627	5.0522
	(0.0089)	(0.0026)	(0.0368)	(0.3461)	(0.0368)	(0.3461)	(0.0318)	(0.7580)	(0.3640)	(0.8371)	(0.4214)	(0.8221)
30	0.8596	0.7880	5.3725	4.7994	5.3725	4.7994	0.5001	0.3978	1.1480	0.9093	5.4785	4.9898
	(0.0842)	(0.1272)	(0.5278)	(0.7310)	(0.5278)	(0.7310)	(0.0176)	(0.1742)	(0.1850)	(0.3925)	(0.7399)	(0.9649)
40	0.9380	0.8658	4.9062	4.6176	4.9062	4.6176	0.5348	0.3755	1.0696	0.9331	5.2572	4.8817
	(0.0394)	(0.0535)	(0.8723)	(0.5136)	(0.8723)	(0.5136)	(0.6472)	(0.0980)	(0.5165)	(0.5284)	(0.2725)	(0.6107)
50	0.9448	0.8721	4.5058	4.3605	4.5058	4.3605	0.5013	0.4101	0.9683	0.8316	5.0809	4.7961
	(0.0374)	(0.0504)	(0.3927)	(0.2764)	(0.3927)	(0.2764)	(0.9867)	(0.2325)	(0.7643)	(0.1129)	(0.7287)	(0.3807)
100	0.9050	0.8296	4.4495	4.4495	4.4495	4.4495	0.4468	0.4239	0.9739	0.9166	4.8808	4.7548
	(0.0606)	(0.0889)	(0.3490)	(0.3577)	(0.3490)	(0.3577)	(0.4733)	(0.3136)	(0.8054)	(0.4335)	(0.6081)	(0.2933)
250	1.5306	1.4456	5.2721	5.2721	5.2721	5.2721	0.4780	0.4430	0.9676	0.9326	4.7564	4.7214
	(0.0001)	(0.0000)	(0.6712)	(0.6685)	(0.6712)	(0.6685)	(0.7707)	(0.4541)	(0.7617)	(0.5305)	(0.2967)	(0.2364)
SMB												
8	0.7052	0.3526	2.1862	0.7052	10.7193	7.6164	0.8730	0.2608	2.2902	0.8730	10.4989	7.7098
	(0.3023)	(0.4313)	(0.0001)	(0.2646)	(0.0000)	(0.0000)	(0.0000)	(0.0014)	(0.0000)	(0.2307)	(0.0000)	(0.0000)
10	0.7768	0.4237	1.8362	0.8475	9.3220	7.6271	0.7825	0.3175	1.8031	0.9413	9.8095	7.8929
	(0.1721)	(0.6841)	(0.0046)	(0.5640)	(0.0000)	(0.0000)	(0.0005)	(0.0151)	(0.0000)	(0.5793)	(0.0000)	(0.0000)
20	0.3556	0.2134	1.3514	1.0669	8.1792	7.5391	0.4541	0.2044	1.3056	0.9310	8.3560	7.6294
	(0.4182)	(0.1276)	(0.2088)	(0.8011)	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.1512)	(0.5150)	(0.0000)	(0.0000)
30	0.2865	0.2149	1.2178	0.9312	8.0229	7.5931	0.3183	0.1705	1.1594	0.9320	8.0586	7.5017
	(0.2188)	(0.1310)	(0.4291)	(0.7962)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.2180)	(0.5217)	(0.0000)	(0.0000)

Table 12 continued

Window	Crisis period						Non-crisis period					
	0.5%		1%		5%		0.5%		1%		5%	
	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA
40	0.2886	0.2886	0.9380	0.8658	7.4315	7.0707	0.3414	0.2390	1.0696	0.9217	7.9882	7.5102
	(0.2254)	(0.2645)	(0.8145)	(0.6156)	(0.0001)	(0.0004)	(0.0253)	(0.0005)	(0.5165)	(0.4608)	(0.0000)	(0.0000)
50	0.2907	0.1453	1.0901	1.0174	7.0494	6.9041	0.3873	0.2962	1.0481	0.9569	7.8606	7.4391
	(0.2322)	(0.0622)	(0.7405)	(0.9482)	(0.0010)	(0.0012)	(0.1191)	(0.0068)	(0.6533)	(0.6851)	(0.0000)	(0.0000)
100	0.4525	0.4525	1.0558	1.0558	7.1644	7.1644	0.3896	0.3323	0.9968	0.9395	7.2296	7.0692
	(0.8031)	(0.8062)	(0.8396)	(0.8382)	(0.0007)	(0.0003)	(0.1280)	(0.0263)	(0.9760)	(0.5700)	(0.0000)	(0.0000)
250	1.1054	1.1054	2.7211	2.7211	9.0136	8.9286	0.3381	0.3148	1.0026	0.9792	6.5983	6.5633
	(0.0112)	(0.0032)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0239)	(0.0150)	(0.9810)	(0.8468)	(0.0000)	(0.0000)
RMW												
8	1.9041	0.4937	3.1030	1.3399	9.7320	7.8279	1.5306	0.6236	2.8005	1.2472	9.6145	6.8821
	(0.0000)	(0.9730)	(0.0000)	(0.1983)	(0.0000)	(0.0000)	(0.0000)	(0.0999)	(0.0000)	(0.0197)	(0.0000)	(0.0000)
10	1.7655	0.7768	2.7542	1.7655	8.8277	7.4153	1.1907	0.5557	2.2908	1.1907	8.9022	6.8723
	(0.0000)	(0.1397)	(0.0000)	(0.0038)	(0.0000)	(0.0000)	(0.0000)	(0.4585)	(0.0000)	(0.0718)	(0.0000)	(0.0000)
20	0.9957	0.7112	1.9915	1.6358	7.5391	6.5434	0.7153	0.5563	1.6349	1.2489	7.8678	6.9936
	(0.0203)	(0.2615)	(0.0010)	(0.0166)	(0.0000)	(0.0079)	(0.1830)	(0.4537)	(0.2651)	(0.0189)	(0.0000)	(0.0000)
30	1.0029	0.7163	1.7192	1.5043	7.3782	6.3754	0.7388	0.5228	1.5117	1.2048	7.3994	6.7402
	(0.0192)	(0.2518)	(0.0143)	(0.0583)	(0.0001)	(0.0184)	(0.7750)	(0.7613)	(0.0937)	(0.0535)	(0.0000)	(0.0000)
40	0.6494	0.5772	1.9481	1.4430	6.9264	6.4214	0.6145	0.5234	1.4679	1.2289	6.9071	6.4747
	(0.4512)	(0.6837)	(0.0017)	(0.0974)	(0.0018)	(0.0152)	(0.1419)	(0.7554)	(0.0000)	(0.0310)	(0.0000)	(0.0000)
50	0.6541	0.5814	1.9622	1.5988	6.6134	6.3953	0.6607	0.5126	1.4582	1.2303	6.8239	6.4821

Table 12 continued

Window	Crisis period						Non-crisis period					
	0.5%		1%		5%		0.5%		1%		5%	
	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA	MA	Adj. MA
	(0.4393)	(0.6686)	(0.0015)	(0.0256)	(0.0087)	(0.0176)	(0.0419)	(0.8666)	(0.0001)	(0.0301)	(0.0000)	(0.0000)
100	0.8296	0.8296	1.7345	1.7345	5.8069	5.7315	0.6531	0.6187	1.3176	1.2374	6.4276	6.3359
	(0.1203)	(0.0889)	(0.0149)	(0.0072)	(0.1882)	(0.2216)	(0.0529)	(0.1159)	(0.0045)	(0.0258)	(0.0000)	(0.0000)
250	1.5306	1.5306	2.2959	2.2959	7.2279	7.2279	0.5712	0.5596	1.1308	1.1308	6.2719	6.2019
	(0.0001)	(0.0000)	(0.0001)	(0.0000)	(0.0010)	(0.0005)	(0.3605)	(0.4341)	(0.2330)	(0.2234)	(0.0000)	(0.0000)

The left panels represent tests over the crisis periods marked by the NBER since 1980 (the last five recessions). For each window and return series, the VaR is estimated i) using the rolling window estimation of the variance (MA); ii) adjusting the VaR from MA to correct for estimation error (Adj. MA. The p values in parentheses are in i) MA, for testing that the violation rate is equal to the theoretical rate using our proposed test; ii) Adj. MA, for testing that the violation rate is equal to the corresponding expected violation rate level using our proposed test

By definition,

$$\begin{aligned}\alpha &= \Pr(-X \geq VaR_\alpha) \\ &= \Pr(X \leq -VaR_\alpha) \\ &= F_x(-VaR_\alpha) \\ &= F_x(E(-\widehat{\upsilon}_\alpha))\end{aligned}$$

And using the assumption that F_x the cdf of X_k is a convex function around $-VaR_\alpha$, we have that $F_x(E(-\widehat{\upsilon}_\alpha)) \leq E(F_x(-\widehat{\upsilon}_\alpha))$ **Q.E.D**

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